
How Architecture Topology Shapes Normalization-Free Network Effectiveness

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Abstract

Dynamic Tanh (DyT) has been proposed as a normalization-free alternative to Layer-Norm in Transformers, but its effectiveness varies across architectures without systematic explanation. We analyze DyT through the lens of information-flow topology—open-loop (attention-only), closed-loop (SSM), and hybrid—and identify two further factors: depth-width ratio (governing tanh compression accumulation) and MLP sub-layers (a compensatory channel). We evaluate five architectures with three seeds on CIFAR-100 (32×32)—DeiT-S, DeiT-T, Vim-T, PlainMamba-T, and RWKV-T—with cross-dataset validation on Tiny-ImageNet (64×64), augmented by single-seed probes (CaiT-XXS24, Sequencer2D-S, RMT-T) and three-seed Swin-T, and single-seed ConvNeXt-T (3-seed re-run pending). Global standard self-attention architectures benefit most: DeiT-S gains +6.90 pp on CIFAR-100 and +6.35 pp on Tiny-ImageNet; DeiT-T shows consistent gains (+3.91 pp on CIFAR-100). The closed-loop Vim-T improves on CIFAR-100 (+3.45 pp) but only marginally on Tiny-ImageNet (+1.40 pp, 3 seeds), with sign reversal across seeds and a supplementary seed exhibiting training collapse. The hybrid RWKV-T *degrades* under DyT (−2.17 pp on CIFAR-100, −2.05 pp on Tiny-ImageNet, 3 seeds each); a targeted ablation shows that the majority of this degradation stems from structural normalization removal, with a smaller but consistent contribution from architecture-specific `key_norm` removal, revealing a distinction between functional and structural normalization roles. RMT-T (VisRetNet), an open-loop retention architecture, suffers catastrophic training collapse under DyT (single seed), demonstrating that the DyT benefit within open-loop designs is specific to global standard self-attention rather than a general property of open-loop information flow. A depth-width matched comparison (DeiT-S vs. PlainMamba-T, both $12L/d=384$, 7.31 pp swing) implicates MLP availability, with Vim-T (closed-loop *with* MLP, +3.45 pp) providing partial disentanglement ($n=1$ ablations). Sensitivity to the DyT initialization scale α is also topology-dependent: DeiT-S is competitive only at $\alpha \leq 0.50$ and collapses at $\alpha \geq 1.0$; RWKV-T remains competitive across $\alpha \in [0.50, 2.00]$; Vim-T converges only at $\alpha=0.50$. These results yield three contributions: (1) the first systematic cross-topology evaluation of DyT, revealing topology-dependent patterns in both effectiveness direction and viable α -range; (2) an empirical characterization of α -deadlock—consistent with known tanh saturation dynamics—identifies a resolution-dependent DyT failure mode; and (3) the RWKV-T `key_norm` decomposition provides a principled distinction between functional and structural normalization. Within the architectures and low-resolution settings tested, DyT effectiveness is topology-dependent; replacing LayerNorm with DyT requires per-architecture validation rather than blanket adoption.

1 Introduction

LayerNorm [Ba et al., 2016] is standard in modern Transformers [Vaswani et al., 2017], but introduces per-token statistical computations. Dynamic Tanh (DyT) [Zhu et al., 2025] replaces LayerNorm with $\gamma \cdot \tanh(\alpha x) + \beta$, eliminating running statistics while maintaining competitive performance on Vision Transformers [Dosovitskiy et al., 2021] and language models.

Yet vision architectures now span diverse *information-flow topologies*: SSMs like Mamba/Vim [Gu and Dao, 2024, Zhu et al., 2024] introduce closed-loop recurrence; hybrids like Vision-RWKV [Duan et al., 2024, Peng et al., 2023] combine linear attention with convolutional priors and specialized normalization (e.g., `key_norm`). Existing DyT evaluations focus on open-loop attention [Zhu et al., 2025]; whether DyT transfers to recurrent or hybrid topologies is uninvestigated. This matters because topology shapes how normalization errors propagate: independently per-layer in open-loop designs, or compounding through recurrent state updates in closed-loop ones.

We address this gap through controlled experiments on CIFAR-100 and Tiny-ImageNet across ten architectures: DeiT-S/T (open-loop) [Touvron et al., 2021a], Vim-T and PlainMamba-T (closed-loop) [Zhu et al., 2024, Yang et al., 2024], RWKV-T (hybrid) [Peng et al., 2023], with probes on CaiT-XXS24 [Touvron et al., 2021b], Sequencer2D-S [Tatsunami and Taki, 2022], RMT-T [Fan et al., 2024], ConvNeXt-T, and Swin-T. Our contributions:

1. **First systematic cross-topology evaluation of DyT.** On CIFAR-100, DyT improves DeiT-S by +6.90 pp (open-loop) and Vim-T by +3.45 pp (closed-loop), but *degrades* RWKV-T by −2.17 pp (hybrid). The benefit is specific to global standard self-attention: CaiT-XXS24 (−12.3 pp), Swin-T (−17.18 pp), ConvNeXt-T (−12.11 pp), and RMT-T (collapse) all degrade. A three-factor analysis (depth–width ratio, MLP availability, topology) refines the lens: DeiT-S vs. PlainMamba-T (both 12L/d=384, 7.31 pp swing) implicates MLP; Vim-T provides partial disentanglement ($n=1$ ablations). α -sweeps reveal topology-specific stability: RWKV-T converges at all five tested values, DeiT-S at $\alpha \leq 0.50$ only, Vim-T at $\alpha=0.5$ only. Cross-dataset validation confirms the pattern (+6.35 pp for DeiT-S on Tiny-ImageNet).
2. **α -deadlock at 224×224 .** On ImageNet-100, all tested α values converge to 48–52% accuracy (18–23 pp below LN), consistent with tanh saturation dynamics [Glorot and Bengio, 2010, Chen et al., 2025] (§K).
3. **Hybrid degradation decomposition.** RWKV-T’s DyT degradation decomposes into structural norms ($\sim 65\%$) and `key_norm` ($\sim 35\%$), indicating broad incompatibility beyond the specialized site.

2 Related Work

2.1 Normalization in Deep Learning

Batch Normalization (BN) [Ioffe and Szegedy, 2015] normalizes activations across the mini-batch dimension but depends on batch statistics, making it ill-suited for small batches or variable-length sequences. LayerNorm (LN) [Ba et al., 2016] normalizes across features within each sample and has become the default in Transformers [Vaswani et al., 2017]. Variants include GroupNorm [Wu and He, 2018] for convolutional architectures and RMSNorm [Zhang and Sennrich, 2019], which removes mean-centering for efficiency. Theoretically, Santurkar et al. [2018] show that normalization smooths the loss landscape rather than reducing internal covariate shift, providing a principled basis for understanding why its removal may destabilize certain architectures.

Beyond activation normalization, Query-Key Normalization [Henry et al., 2020] stabilizes attention logits by constraining query and key magnitudes—illustrating that normalization serves multiple distinct functional roles, complicating uniform replacement.

Pre-norm placement yields more stable gradients than post-norm [Xiong et al., 2020], while DeepNet [Wang et al., 2024] stabilizes Post-LN via residual scaling. Both findings underscore that normalization effectiveness is architecture-dependent—a theme central to our investigation.

2.2 Normalization-Free Networks

In the convolutional setting, Fixup [Zhang et al., 2019] replaces BN via careful initialization in ResNets [He et al., 2016], and NFNet [Brock et al., 2021] combine signal propagation analysis with Adaptive Gradient Clipping. Both are tailored to CNNs and do not transfer to Transformers, where unbounded attention scores and residual stream activations pose distinct challenges. ReZero [Bachlechner et al., 2021] bridges this gap by initializing residual branch weights to zero in Transformers, enabling norm-free training through signal propagation control rather than activation normalization.

Dynamic Tanh (DyT) [Zhu et al., 2025] takes a different approach: replacing LN with a learnable bounded nonlinearity $\gamma \cdot \tanh(\alpha \cdot x) + \beta$ as a *post-hoc substitution* that keeps training procedures unchanged. The tanh saturation connects to classical vanishing gradient findings [Glorot and Bengio, 2010]. Concurrently, Chen et al. [2025] extend DyT with designs that mitigate saturation. Both were validated primarily on global self-attention architectures, leaving applicability to other topologies unexplored.

These efforts focus on *whether* normalization can be removed; our work asks *when*—under what topology conditions a representative LN replacement succeeds or fails.

2.3 Vision Architecture Topologies

Modern vision architectures span a spectrum of information-flow topologies. ViT [Dosovitskiy et al., 2021] and DeiT [Touvron et al., 2021a] process tokens through stacked global self-attention in a purely feedforward (*open-loop*) manner. CaiT [Touvron et al., 2021b] modifies this pattern by separating self-attention and class-attention into distinct stages, introducing additional normalization sites at the class-token interaction boundary. Swin [Liu et al., 2021] restricts attention to local windows with hierarchical pooling, and ConvNeXt [Liu et al., 2022] modernizes CNNs with large-kernel depthwise convolutions and LN—both producing hierarchical multi-resolution feature maps rather than a columnar token sequence.

State-space models introduce recurrent dynamics. S4 [Gu et al., 2022] provides the continuous-time foundation; Mamba [Gu and Dao, 2024] adds input-dependent selection with a hidden state updated recurrently. Vision Mamba (Vim) [Zhu et al., 2024] adapts this to image patches with bidirectional scanning, creating a *closed-loop* feedback path. PlainMamba [Yang et al., 2024] further simplifies the architecture by removing MLP blocks entirely, making normalization the sole inter-layer mechanism regulating signal magnitude before recurrent state transitions. Sequencer2D [Tatsunami and Taki, 2022] applies bidirectional LSTMs along both spatial axes. xLSTM [Beck et al., 2024], Mamba-2 [Dao and Gu, 2024], and VMamba [Liu et al., 2024] provide additional closed-loop designs.

Hybrid architectures lie between these extremes. Vision-RWKV [Duan et al., 2024], based on RWKV [Peng et al., 2023], interleaves linear attention with spatial convolution priors and employs functional normalization (`key_norm`) beyond generic activation stabilization; RWKV-5/6 [Peng et al., 2024] extend this with matrix-valued states. RetNet [Sun et al., 2023] and RMT [Fan et al., 2024] similarly blend parallel attention with recurrent computation.

This topology distinction is central to our analysis: prior normalization replacement work treats architectures as interchangeable targets, whereas our results reveal information-flow topology as a primary factor determining replacement feasibility.

2.4 Architecture-Dependent Normalization

The functional role of normalization varies across architecture families. In open-loop Transformers, LN stabilizes per-token activations in the residual stream. In CaiT [Touvron et al., 2021b], additional normalization at the class-attention boundary mediates information flow between self-attention and class-attention stages—a structural role absent from standard ViTs. In closed-loop SSMs [Gu and Dao, 2024, Zhu et al., 2024], LN additionally constrains recurrent state dynamics; PlainMamba [Yang et al., 2024] makes this coupling especially tight by removing MLP blocks, so normalization directly gates state transitions without an intermediate nonlinear bottleneck. In hybrids such as Vision-RWKV [Duan et al., 2024], certain normalization layers (e.g., `key_norm`) serve structural roles [Henry et al., 2020] that generic replacements cannot replicate. Hierarchical designs—Swin [Liu et al., 2021]

and ConvNeXt [Liu et al., 2022]—add further complexity: LN must stabilize activations across resolution stages where feature statistics change abruptly at downsampling boundaries. ConvNeXt V2 [Woo et al., 2023] further illustrates this architecture-specificity by introducing Global Response Normalization (GRN), a channel-wise normalization tailored to convolutional feature maps that standard LN or its replacements cannot subsume.

No prior work has systematically investigated how architecture topology determines normalization replacement feasibility.

2.5 DyT Across Architectures

Zhu et al. [2025] and Chen et al. [2025] evaluated DyT on ViT, DiT, and LLMs—all sharing global self-attention. Whether DyT transfers to closed-loop, hierarchical, or hybrid architectures has not been investigated.

We fill this gap by evaluating DyT across ten vision architectures spanning open-loop, closed-loop, hierarchical, and hybrid topologies on CIFAR-100 and Tiny-ImageNet, providing the first systematic evidence that architecture topology is a primary determinant of whether LN can be safely replaced. Only global standard self-attention architectures (DeiT-S/T) consistently benefit; hierarchical, class-attention, recurrent, and hybrid designs degrade or collapse.

3 Method

3.1 Architecture Taxonomy

We select four architectures spanning three information-flow topologies, targeting the *pre-norm residual pattern* with token-wise normalization—shared by standard Transformers, SSMs, and linear attention variants—but excluding hierarchical CNNs where normalization manages cross-stage scale transitions. All models process 32×32 images patchified into token sequences.

Open-loop: DeiT-S and DeiT-T. DeiT [Touvron et al., 2021a] computes multi-head self-attention over all token pairs followed by a feed-forward network, with no recurrent state (*open-loop*). We include DeiT-S (~ 22 M) and DeiT-T (~ 5.4 M) to disentangle topology from capacity effects.

Closed-loop: Vim-T. Vision Mamba (Vim) [Zhu et al., 2024] applies Mamba’s [Gu and Dao, 2024] Selective State Space Model to vision, maintaining a hidden state updated recurrently: $h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$, $y_t = C_t h_t$, forming a *closed-loop* feedback path (~ 22 M parameters).

Hybrid: RWKV-T. Vision-RWKV [Duan et al., 2024], based on RWKV [Peng et al., 2023], combines convolutional priors (q_shift) with bidirectional WKV attention. Its distinctive `key_norm` bounds key magnitudes before attention computation. This *hybrid* is neither purely feedforward nor recurrent (~ 5.8 M parameters).

3.2 DyT Replacement Protocol

DyT [Zhu et al., 2025] replaces LayerNorm with a learnable bounded nonlinearity:

$$\text{DyT}(x) = \gamma \cdot \tanh(\alpha x) + \beta, \quad (1)$$

where α controls saturation and γ, β are affine parameters. Unlike LN, DyT applies a pointwise nonlinearity without per-token statistics.

Universal replacement. We traverse the full module tree of each architecture and substitute every LayerNorm with DyT (initialized $\alpha = 0.5$), ensuring consistent protocol across all ten architectures ($< 0.1\%$ parameter change).

Key-norm ablation. RWKV-T’s key_norm serves a structural role distinct from standard pre-/post-block norms. We introduce selective exclusion: three configurations—full LN baseline, full DyT, and DyT with key_norm preserved as LN—isolate the key normalization contribution.

3.3 Training Setup

Datasets. Primary experiments use CIFAR-100 (100 classes, 50K training images, 32×32 , standard test split). Cross-dataset validation uses Tiny-ImageNet (200 classes, 100K images, 64×64 resized to 32×32).

Augmentation and regularization. Mixup [Zhang et al., 2018] ($\alpha = 0.8$), cutmix [Yun et al., 2019] ($\alpha = 1.0$), label smoothing [Szegedy et al., 2016] ($\epsilon = 0.1$), and stochastic depth [Huang et al., 2016] (rate = 0.1), held constant across all architectures.

Optimization. All models trained from scratch with AdamW [Loshchilov and Hutter, 2019] (lr = 10^{-3} , weight decay = 0.05), cosine schedule [Loshchilov and Hutter, 2017] without warmup, 100 epochs, batch size 128, FP32, patch size 4 (64 tokens), on a single NVIDIA RTX 5090. No architecture-specific tuning is applied.

Evaluation and controls. Best top-1 test accuracy averaged over three seeds (42, 123, 0). Two parameter-matched pairs—DeiT-S/Vim-T (~ 22 M) and DeiT-T/RWKV-T (~ 5.5 M)—isolate topology from scale.

Reproducibility. Implementation based on PyTorch and timm. Code available upon publication.

4 Results

4.1 Architecture Topology and DyT Effectiveness at Low Resolution

Table 1 summarizes the effect of replacing LayerNorm with DyT across ten architectures spanning three topology categories. All models are trained on CIFAR-100 for 100 epochs with identical optimization settings per architecture; we report test accuracy averaged over three seeds for eight architectures (DeiT-S/T, Vim-T, PlainMamba-T, RWKV-T, CaiT-XXS24, ConvNeXt-T, Swin-T) and single-seed results for the auxiliary probes (RMT-T, Sequencer2D-S; marked †).

Three outcome groups. The ten architectures partition into: positive ($\Delta \geq +3$ pp)—DeiT-S, DeiT-T, Vim-T; near-zero—PlainMamba-T; negative or collapse—RWKV-T, CaiT-XXS24, ConvNeXt-T, Swin-T, RMT-T, Sequencer2D-S. Within open-loop, only global standard self-attention at moderate depth benefits; closed-loop SSMs split (Vim-T positive, PlainMamba-T near-zero); the hybrid RWKV-T degrades uniformly. The mechanistic account is deferred to §5.6.

Global standard self-attention: positive at moderate depth. DeiT-S gains +6.90 pp (58.0% $\rightarrow$ 64.9%) and DeiT-T gains +3.91 pp (56.9% $\rightarrow$ 60.8%); both are direction-consistent across all seeds. DyT also reduces DeiT-S inter-seed std from 1.6 to 0.6. However, the open-loop label alone does not predict positive Δ : CaiT-XXS24 (24 layers) drops 12.34 pp, ConvNeXt-T drops 12.11 pp, and RMT-T suffers catastrophic collapse. A targeted CaiT ablation (DyT only on SA norms, FFN retains LN) yields 49.9%, indistinguishable from full-DyT (49.5%), localizing the failure to the deep SA stack. Swin-T shows the largest deficit among converging architectures ($\Delta = -17.18$ pp, 3-seed mean), though this is resolution/architecture-confounded: Zhu et al. [2025] reported positive Swin-T results at 224×224 . Even the best $\alpha=1.0$ remains 9.8 pp below LN, indicating the deficit is structural rather than an initialization artifact (Table 14).

Closed-loop SSM: split outcome. Vim-T gains +3.45 pp (61.08% $\rightarrow$ 64.53%), but converges only at $\alpha=0.5$ (§4.6). PlainMamba-T is near-zero (-0.41 ± 0.18 pp). Although the two SSMs differ substantially in capacity (Vim-T 22M vs. PlainMamba-T 12.1M, $\sim 2\times$), capacity does not predict DyT effectiveness across architectures: DeiT-S (22M) gains +6.90 pp while ConvNeXt-T (27.9M) degrades -12.11 pp, ruling out parameter count as the explanation for this divergence. The two SSM

Table 1: Effect of replacing LayerNorm with DyT across ten architectures spanning three topology categories on CIFAR-100 test accuracy (%). Mean $\pm$ std over 3 seeds; entries marked $\dagger$ are single-seed probes. Δ : DyT – LN (percentage points), computed from unrounded means. Best α from the sensitivity sweep (§4.6); “—” indicates α was not swept for that architecture. Rows are grouped by Δ sign (positive / near-zero / negative). $\ddagger$ Swin-T uses architecture-specific hyperparameters ($\text{lr}=5\times 10^{-4}$, 5-epoch warmup, gradient clipping=5.0) as standard DeiT recipes do not converge for this architecture at 32×32 . *CaiT-XXS24 ablation (basic recipe)*: replacing only the self-attention norms with DyT (FFN remains LN) yields 49.9% ($\Delta = -8.0$ pp vs. basic-recipe LN of 57.9%), essentially indistinguishable from the basic-recipe full-DyT result (49.5%). $\S$ ConvNeXt-T LN exhibits high inter-seed variance ($\sigma=2.45$ pp; seeds: 55.2, 57.5, 52.6%), substantially above other 3-seed architectures ($\sigma \leq 1.55$ pp).

Architecture	Topology	Params	LN Acc (%)	DyT Acc (%)	Δ (pp)	Best α
DeiT-S	Open-loop (attn)	22M	58.0 ± 1.6	64.9 ± 0.6	+6.90	0.50
DeiT-T	Open-loop (attn)	5.4M	56.9 ± 0.1	60.8 ± 0.6	+3.91	—
Vim-T	Closed-loop (SSM)	22M	61.1 ± 0.6	64.5 ± 0.8	+3.45	0.50
PlainMamba-T	Closed-loop (SSM)	12.1M	74.2 ± 0.1	73.8 ± 0.2	−0.41	—
RWKV-T	Hybrid	5.8M	74.6 ± 0.2	72.5 ± 0.2	−2.17	1.00
CaiT-XXS24	Open-loop (deep attn)	11.6M	68.9 ± 0.1	56.6 ± 0.4	−12.34	2.00
ConvNeXt-T $\S$	Open-loop (CNN)	27.9M	55.1 ± 2.5	43.0 ± 0.7	−12.11	—
Swin-T $\ddagger$	Open-loop (windowed attn)	27.6M	64.0 ± 0.1	46.8 ± 0.3	−17.18	1.0
RMT-T $\dagger$	Open-loop (retention)	13.4M	56.4	collapse (30.1)	—	—
Sequencer2D-S $\dagger$	Closed-loop (BiLSTM)	27M	61.1	collapse (5.2)	—	—

variants differ in two architectural respects: (i) scan direction—Vim-T uses bidirectional Mamba scanning [Zhu et al., 2024], where each token aggregates context from the entire sequence (analogous in scope to global self-attention), while PlainMamba-T uses unidirectional scanning with learned directional biases [Yang et al., 2024]; (ii) MLP availability—Vim-T includes MLP sub-layers that PlainMamba-T lacks. Both factors are consistent with the broader pattern: architectures enabling global information aggregation benefit from DyT, while those restricted to partial-sequence processing do not (§5.3). Controlled ablations (§5.3) partially disentangle these: MLP removal is necessary for Vim-T’s benefit, but grafting MLP onto PlainMamba-T does not confer benefit ($\Delta = -1.10$ pp), implicating scan direction as an independent factor.

Of the two factors, MLP availability produces the larger quantitative effect—a 14 pp swing upon removal (Table 3)—but its contribution is fully captured by the controlled ablation. Scan direction, by contrast, lacks a direct ablation (one cannot “remove” bidirectionality without changing the architecture fundamentally) and therefore requires the mechanistic argument we develop below.

The divergence between these two SSM subtypes is consistent with a normalization-dependency account tied to information flow symmetry. Vim-T’s bidirectional Mamba scanning [Zhu et al., 2024] processes each token through both forward and backward state transitions, producing representations that aggregate context from the full input sequence—resembling the global information flow of standard self-attention. Under such symmetric aggregation, the full-sequence statistics that LayerNorm would compute are implicitly encoded in the activations, allowing DyT’s learnable $\tanh(\alpha x)$ channel scaling to serve as a functional substitute. PlainMamba-T’s unidirectional scanning with learned directional biases [Yang et al., 2024], by contrast, yields position-dependent representations shaped by partial-sequence context and direction-specific scaling. This suggests that LN’s explicit statistical normalization remains functionally non-redundant under asymmetric information flow: directional biases create position-varying activation distributions that $\tanh$ scaling—a position-agnostic element-wise operation—cannot accommodate. The near-zero Δ (−0.41 pp) for PlainMamba-T, rather than catastrophic failure, is consistent with its directional biases providing partial implicit normalization that buffers against—but does not benefit from—the LN-to-DyT substitution. However, this analogy between bidirectional scanning and global self-attention has limits: the α -sensitivity sweep (§4.6) reveals that Vim-T converges at only one tested α value (0.5), whereas DeiT-S—also

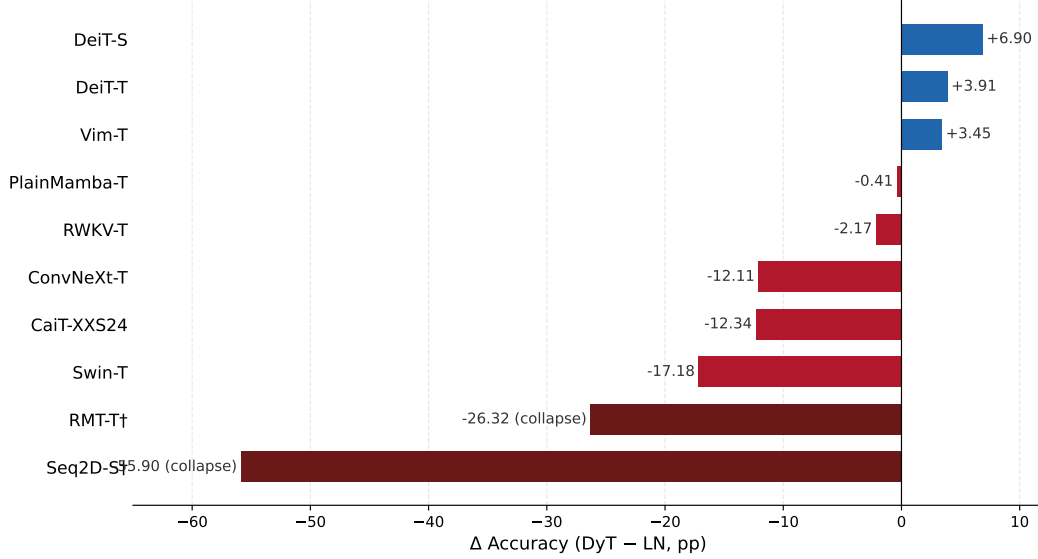


Figure 1: Per-model Δ accuracy (DyT – LayerNorm) on CIFAR-100 across ten models. Positive values (blue) indicate DyT improvement; negative bars (red) indicate degradation, with Swin-T (-17.18 pp) showing the largest deficit among converging architectures.

a global-aggregation architecture—converges at three ($\alpha \leq 0.50$). This narrower stability window suggests that SSM’s recurrent state compression imposes tighter numerical conditioning on activations than attention’s explicit pairwise computation, making the LN-to-DyT substitution more fragile even when the *scope* of aggregation is comparably global. Aggregation scope thus does not fully determine normalization sensitivity; the *mechanism* of aggregation—parallel pairwise (SA) vs. sequential recurrent (SSM)—introduces an independent conditioning axis.

The hybrid RWKV-T is hurt by DyT. RWKV-T drops from 74.63% to 72.45% (-2.17 pp); at comparable scale, DeiT-T gains $+3.91$ pp, ruling out scale. The decomposition (§4.3) attributes $\sim 65\%$ to structural norms and $\sim 35\%$ to `key_norm`.

DeiT-S with DyT reaches 50% of best accuracy in half the epochs vs. LN (10.0 ± 1.0 vs. 19.0 ± 2.6), though this advantage is not consistent across architectures (Appendix A). The topology-dependent patterns extend to all four Tiny-ImageNet combinations (§4.5).

4.2 Disentangling Topology from Scale

Two parameter-matched pairs isolate topology from scale. At ~ 22 M, DeiT-S (open-loop) gains $+6.90$ pp while Vim-T (closed-loop) gains $+3.45$ pp—both benefit, but topology modulates magnitude. At ~ 5.5 M, DeiT-T ($+3.91$ pp) and RWKV-T (-2.17 pp) show a sign reversal at matched capacity; topology is the only variable that differs. Both DeiT variants benefit direction-consistently across a $4\times$ parameter range, though two scale points are insufficient for a scaling relationship.

4.3 Decomposing the RWKV-T Performance Gap

Given that RWKV-T is the only multi-seed core architecture where DyT degrades performance, we perform a targeted ablation to identify which normalization layers contribute to this degradation. Specifically, we compare three configurations: (1) the full LayerNorm baseline, (2) DyT replacement of all norms *except* `key_norm`, and (3) full DyT replacement including `key_norm`. Results are reported across three seeds (42, 123, 0).

Table 2 shows the 2.17 ± 0.14 pp degradation decomposes into two components: preserving `key_norm` recovers 0.77 ± 0.33 pp ($\sim 35\%$), while structural norms account for 1.41 ± 0.46 pp ($\sim 65\%$). The `key_norm` contribution varies across seeds (42: 40%; 123: 52%; 0: 17%), but structural norms dominate in two of three.

Table 2: Decomposition of the DyT performance gap in RWKV-T on CIFAR-100. DyT (excl. key_norm) replaces all norms except key_norm with DyT. Structural norms account for $\sim 65\%$ and key_norm for $\sim 35\%$ of the total degradation (3-seed mean), though the relative contribution varies across seeds (structural: 48–83%).

Configuration	Seed	Acc (%)	Δ vs Full LN (pp)	Gap Explained
Full LN (baseline)	42	74.46	—	—
DyT (excl. key_norm)	42	73.13	−1.33	Structural: 60%
Full DyT	42	72.26	−2.20	+ key_norm: 40%
Full LN (baseline)	123	74.52	—	—
DyT (excl. key_norm)	123	73.54	−0.98	Structural: 48%
Full DyT	123	72.50	−2.02	+ key_norm: 52%
Full LN (baseline)	0	74.90	—	—
DyT (excl. key_norm)	0	73.00	−1.90	Structural: 83%
Full DyT	0	72.60	−2.30	+ key_norm: 17%
<i>3-seed mean $\pm$ std</i>			-2.17 ± 0.14	Structural: $64 \pm 17\%$ / key_norm: $36 \pm 17\%$

Table 3: MLP compensation ablation on CIFAR-100 (seed = 42). Removing MLP from Vim-T reverses the DyT advantage; adding MLP to PlainMamba-T does not confer one.

Architecture	Variant	DyT (%)	LN (%)	Δ (pp)
Vim-T	Full (with MLP)	63.60	61.30	+2.30
Vim-T	−MLP	48.54	60.30	−11.76
PlainMamba-T	Original (no MLP)	73.60	74.13	−0.53
PlainMamba-T	+MLP	63.40	64.50	−1.10

This reveals two distinct incompatibility sources: key_norm enforces a magnitude invariant that DyT cannot replicate, while the broader structural gap indicates DyT is incompatible with the hybrid architecture beyond the specialized key_norm site. Selective replacement preserving architecture-specific functional norms may mitigate degradation.

4.4 MLP Compensation Ablation

To isolate the role of MLP sub-layers in mediating DyT effectiveness within closed-loop SSMs, we perform bidirectional ablations on two architectures: Vim-T (which natively includes MLP) and PlainMamba-T (which does not). All results are single-seed (seed = 42).

Table 3 shows MLP is necessary but not sufficient for DyT effectiveness in closed-loop SSMs. Removing MLP from Vim-T flips the DyT effect from +2.30 pp to −11.76 pp (14.06 pp swing), while LN degrades by only 1.00 pp. Conversely, grafting MLP onto PlainMamba-T does not confer DyT tolerance ($\Delta = -1.10$ pp), with both conditions suffering ~ 10 pp degradation. MLP compensation requires a compatible SSM formulation; Vim-T’s bidirectional scan and dual-branch structure appear jointly necessary (§5.3).

4.5 Cross-Dataset Validation

To assess whether the topology-dependent pattern observed on CIFAR-100 generalizes beyond a single dataset, we evaluate the core four multi-seed architectures on Tiny-ImageNet (200 classes, 64×64 images resized to 32×32 , 100K training images).

Table 4 confirms the topology-dependent pattern on Tiny-ImageNet: DeiT-S maintains +6.35 pp, DeiT-T shows attenuated but direction-consistent +0.69 pp (3 seeds), Vim-T is marginal (+1.40 pp, with seed-dependent sign reversal¹; Appendix M), and RWKV-T degrades consistently (−2.05 pp).

¹Vim-T mean is +1.40 pp, but seed 123 shows −1.19 pp; remaining seeds: +4.40 and +1.00 pp.

Table 4: Cross-dataset validation on Tiny-ImageNet. All architectures averaged over 3 seeds (42, 123, 0). CIFAR-100 Δ shown for comparison. [†]Per-seed signs disagree (seed 42: +4.40 pp, seed 123: −1.19 pp, seed 0: +1.00 pp).

Architecture	LN Acc (%)	DyT Acc (%)	Δ (pp)	CIFAR-100 Δ
DeiT-S	39.80 ± 0.72	46.15 ± 0.26	+6.35	+6.90
DeiT-T	41.40 ± 0.26	42.09 ± 0.34	+0.69	+3.91
Vim-T	43.13 ± 2.03	44.54 ± 2.22	+1.40 [†]	+3.45
RWKV-T	54.62 ± 0.33	52.57 ± 0.91	−2.05	−2.17

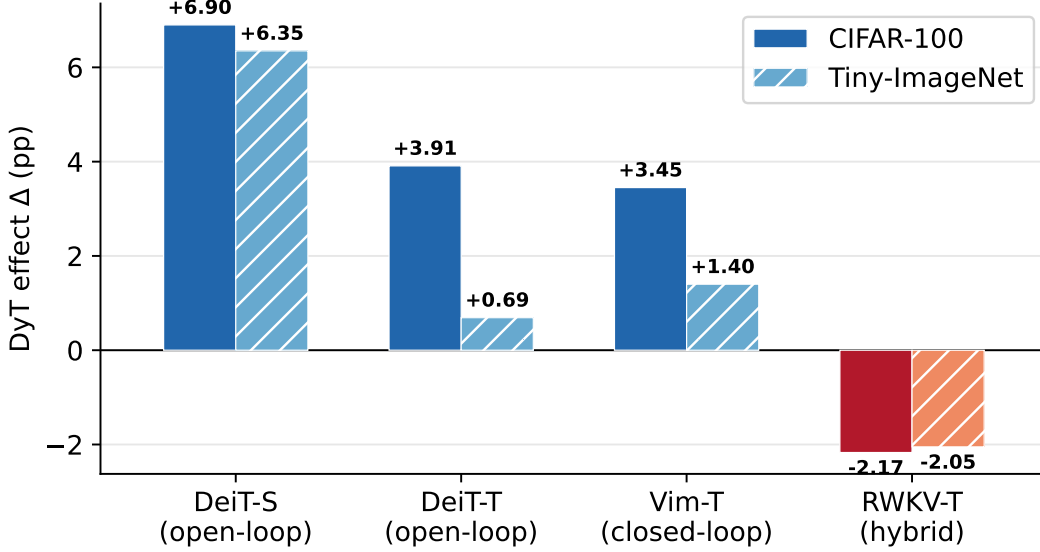


Figure 2: DyT effect (Δ accuracy) on CIFAR-100 vs. Tiny-ImageNet. The topology-dependent pattern—global standard self-attention benefits, hybrid hurts—replicates across both datasets. DeiT-T’s Tiny-ImageNet gain (+0.69 pp, 3 seeds) is substantially attenuated from its CIFAR-100 result (+3.91 pp).

Both datasets use 32×32 inputs; whether the pattern persists at higher resolutions remains open.
Per-seed results are reported in Appendix M.

4.6 Sensitivity to Initialization Scale (α)

The full α -sensitivity sweep (Appendix N, Table 14) reveals topology-dependent stability windows: Vim-T converges only at $\alpha=0.50$; DeiT-S at $\alpha \leq 0.50$; RWKV-T at all tested values, largely closing the DyT-LN gap at $\alpha=1.0$. Swin-T and CaiT-XXS24 converge broadly but remain ≥ 6.6 pp below LN at their best α , confirming structural rather than initialization-related deficits.

α -tuning as topology-dependent phenomenon. Table 1 reports results at the default $\alpha=0.50$ [Zhu et al., 2025]. That RWKV-T’s gap closes at $\alpha=1.0$ does not undermine the taxonomy; the *need* for per-architecture α tuning is itself topology-dependent. Thus α -sensitivity is a complementary dimension: topology is associated with both the *direction* of DyT’s effect and the *stability range* of its initialization scale.

5 Discussion

The topology-dependent pattern in Section 4 invites a binary classification, but this is incomplete: within open-loop, CaiT-XXS24 (−12.34 pp), Swin-T (−17.18 pp), ConvNeXt-T (−12.11 pp), and RMT-T (collapse) all degrade; within closed-loop, Vim-T (+3.45 pp) and PlainMamba-T (−0.41 pp)

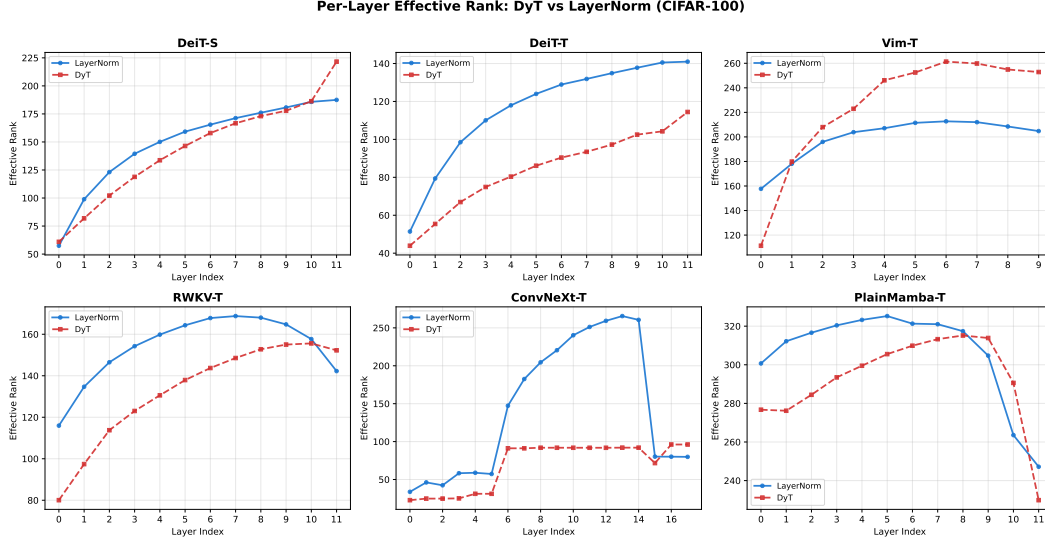


Figure 3: Effective rank change (DyT vs. LN) across architectures. Negative values indicate representational compression under DyT.

disagree. We identify three interacting factors: (1) *depth–width ratio*, governing tanh compression accumulation; (2) *compensatory pathways*, particularly MLP sub-layers; and (3) *topology-specific propagation*. Six of ten architectures are consistent with this framework; ConvNeXt-T, Swin-T, RWKV-T, and RMT-T require architecture-specific considerations. This analysis is empirical and does not constitute formal proof.

5.1 Breaking the Binary: CaiT as Counter-Evidence

CaiT-XXS24 refutes topology sufficiency: DyT degrades accuracy from 68.93% to 56.59% ($\Delta = -12.34$ pp, 3-seed mean, augmented recipe) despite sharing DeiT’s open-loop topology. CaiT differs in depth–width: 24 SA blocks at $d=192$ vs. DeiT-S’s 12 at $d=384$. A targeted ablation applying DyT only to the SA blocks (preserving LN in class-attention) yields 49.9% ($\Delta = -8.0$ pp), recovering only 0.4 pp vs. full-DyT (49.5%)—confirming the deep SA stack as the degradation source. Together with ConvNeXt-T (-12.11 pp; Appendix E), Swin-T (-17.18 pp), and RMT-T (collapse), these four open-loop counter-examples establish that DyT benefits are specific to global standard self-attention at moderate depth.

5.2 Factor 1: Depth–Width Ratio and Compression Accumulation

DyT’s $\tanh(\alpha x)$ imposes per-layer compression whose cumulative effect scales with depth and inversely with representational width. Our architecture set provides three partially disentangled comparisons that isolate the contributions of width, depth, and topology (Figures 3 and 4; Table 5).

Width effect (matched depth). DeiT-T and DeiT-S share the same depth (12L) but differ in width ($d=192$ vs. $d=384$). At matched depth, the wider DeiT-S gains +6.90 pp while the narrower DeiT-T gains +3.91 pp, with effective rank reductions of -3.8% and -27.7% respectively. Doubling width roughly doubles the DyT benefit, consistent with wider representations providing a buffer against tanh compression.

Depth effect (matched width). DeiT-T and CaiT-XXS24 share the same width ($d=192$) but differ in depth (12L vs. 24+2L). At matched width, increasing depth from 12 to 26 layers reverses the DyT effect from +3.91 pp to -12.0 pp (single seed, augmented recipe). Our SA-only ablation (§5.1) confirms that the 24-layer SA stack—not the class-attention mechanism—drives CaiT’s degradation.

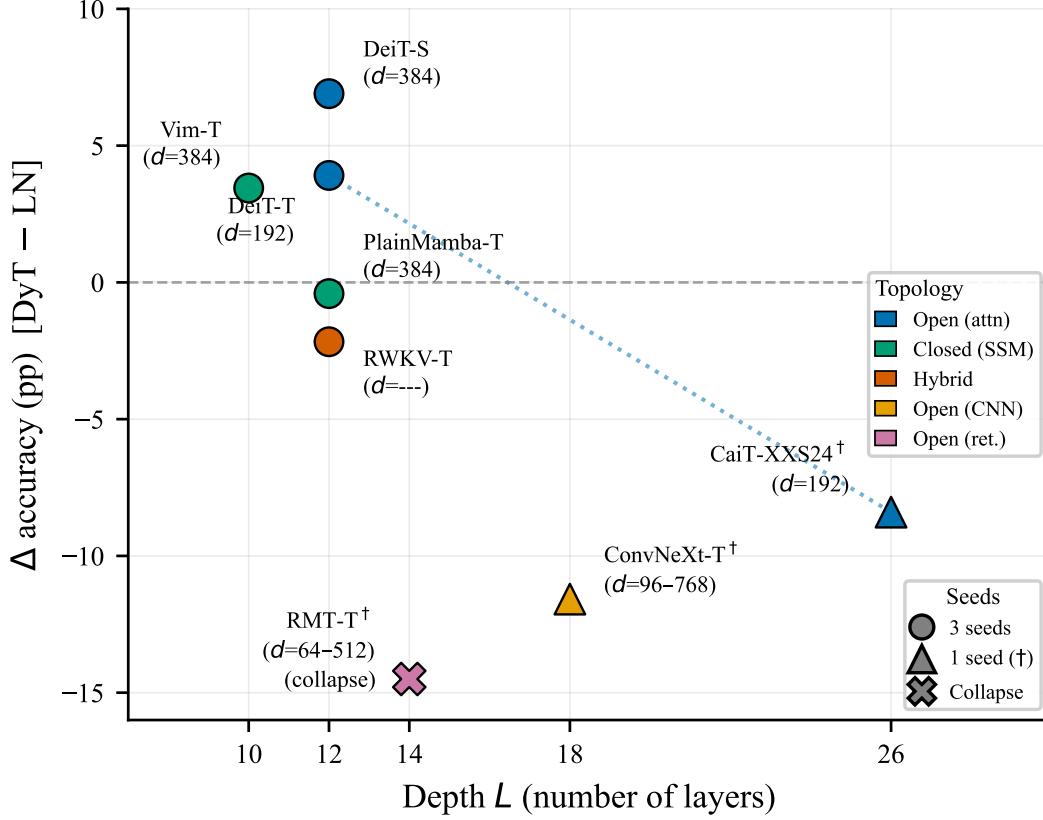


Figure 4: Depth vs. DyT Δ accuracy on CIFAR-100. Color encodes topology; marker shape encodes seed count. The dotted line connects $d=192$ open-loop attention points (DeiT-T $\rightarrow$ CaiT-XXS24), illustrating the depth- Δ gradient. RMT-T (collapse) is clipped to the visible range.

Width provides a representational buffer that $d=192$ sustains at 12 layers of compression but exhausts under 26 layers.

Topology and MLP (matched depth-width). DeiT-S and PlainMamba-T are exactly matched at $12L/d=384$ yet show a 7.31 pp swing (+6.90 vs. -0.41). Vim-T (closed-loop with MLP, +3.45 pp) partially disentangles the confound: closed-loop topology alone does not preclude DyT benefit, implicating MLP availability as a contributing factor. However, scan direction and direction bias also differ between Vim-T and PlainMamba-T; with $n=1$ ablations, the dominant factor remains undetermined.

5.3 Factor 2: Compensatory Pathways — The MLP Channel

MLP sub-layers serve as partial compensatory channels for DyT-induced compression. We establish this through bidirectional ablations (Table 3).

Necessity. Removing MLP from Vim-T flips the DyT effect from +2.30 pp to -11.76 pp (seed = 42), a 14.06 pp swing, while the LN control degrades by only 1.00 pp—confirming the interaction is DyT-specific. MLP activations show $\sim 30\times$ output compression under DyT (std 0.027–0.059 vs. 0.84–1.99), yet Vim-T’s bidirectional SSM maintains representational diversity until MLP is removed.

Insufficiency. Grafting MLP onto PlainMamba-T does not confer DyT tolerance ($\Delta = -1.10$ pp), with both conditions suffering ~ 10 pp degradation. MLP compensation requires a compatible SSM formulation—Vim-T’s bidirectional scan and dual-branch structure appear jointly necessary.

Scan direction and information aggregation. Beyond MLP availability, Vim-T and PlainMamba-T differ fundamentally in how they aggregate information across the input sequence. Vim-T employs bidirectional Mamba scanning [Zhu et al., 2024], where each token simultaneously receives forward and backward context—producing representations that aggregate information from the entire input sequence, achieving a form of global context integration distinct from—but comparable in scope to—self-attention. PlainMamba-T uses unidirectional scanning with learned directional biases [Yang et al., 2024], restricting each position to a single-direction context. This architectural distinction maps onto the broader pattern in Table 1: architectures where every token aggregates information globally (DeiT-S/T, Vim-T) show positive DyT improvements on CIFAR-100 (direction-consistent across all seeds), while those relying on local or sequential processing degrade—though to varying degrees: PlainMamba-T (−0.41 pp) is practically neutral, RWKV-T (−2.17 pp) shows mild degradation, and Swin-T (−17.18 pp) and ConvNeXt-T (−12.11 pp) suffer catastrophic failure. Notably, RWKV-T and PlainMamba-T are both unidirectional sequential architectures, yet differ by $5.3\times$ in Δ . Decomposing RWKV-T’s deficit via `key_norm` ablation (Table 2) into structural ($\sim 65\%$, ≈ -1.41 pp) and `key_norm`-specific ($\sim 35\%$) components brings the structural component closer to PlainMamba-T’s −0.41 pp; the residual gap is attributable to differences in token-mixing—RWKV-T’s WKV attention performs data-dependent weighted averaging, whereas PlainMamba-T’s SSM recurrence applies fixed-structure state transitions. The magnitude gap between partial-sequence architectures—catastrophic for windowed/local models (Swin-T, ConvNeXt-T) vs. mild for sequential SSMs—is further explained by α dead-lock (§K): windowed and local architectures not only lack full-sequence statistics but also trigger $\tanh$ saturation from high-resolution patch embeddings, producing catastrophic failure rather than gradual degradation.

We hypothesize that DyT’s learnable $\tanh(\alpha x)$ can substitute for LN’s statistical normalization when the preceding layer produces globally-aggregated representations: the full-sequence statistics that LN would compute are implicitly available in the activations. Under unidirectional or windowed processing, activations reflect partial-sequence statistics, making LN’s explicit normalization non-redundant. We note that this hypothesis addresses LN’s role in aggregating cross-token statistics; regarding LN’s separate contribution to gradient normalization, a functional substitution relationship exists with DyT’s $\tanh$ -induced implicit gradient clipping: in open-loop architectures, this clipping suffices to replace LN’s gradient normalization, consistent with DeiT-S/T showing positive Δ under DyT; in closed-loop architectures with recurrent gradient flow, the substitution may be insufficient as gradient attenuation compounds across sequential state updates. This is consistent with Vim-T’s positive Δ (+3.45 pp) despite being a closed-loop SSM, and PlainMamba-T’s near-zero Δ (−0.41 pp) despite sharing the same topology category. While Vim-T and PlainMamba-T differ in other respects—including absolute accuracy level, model capacity, sensitivity to the DyT scaling parameter α , and PlainMamba-T’s learned directional biases [Yang et al., 2024] which may confer implicit position-aware scaling that increases robustness to normalization replacement—the contrast in their scan strategies provides a suggestive, if not conclusive, signal. A controlled experiment varying only scan direction (bidirectional $\rightarrow$ unidirectional in Vim-T) would be needed to confirm this hypothesis; however, such modifications require non-trivial changes to the Vim codebase’s scan infrastructure, potentially introducing uncontrolled architectural confounds that would compromise the controlled comparison. As a partial substitute, we ablate PlainMamba-T’s directional biases (removing the learned direction-specific scaling while retaining unidirectional scan): the DyT deficit increases modestly from −0.41 pp to −0.87 pp (single seed), consistent with directional biases providing partial implicit normalization but not demonstrating that adding bidirectionality would confer positive Δ .

Coverage of SSM variants. Our SSM analysis covers two architectures that represent the extremes of the scan-direction axis: fully bidirectional (Vim-T) and fully unidirectional (PlainMamba-T). Other Vision SSM variants—including VMamba [Liu et al., 2024] with 4-directional cross-scan and Mamba-2 [Dao and Gu, 2024] with structured state space duality—were excluded because (i) VMamba’s cross-scan achieves effectively global context aggregation, placing it on the same side as Vim-T under our framework (predicted $\Delta > 0$) and thus providing a weaker test than the existing bidirectional/unidirectional contrast; and (ii) these variants introduce additional mechanisms (2D-selective scan, hardware-aware SSD kernels) that would confound the information-flow variable.

Table 5: Three-factor analysis of DyT effectiveness. [†]Single seed. [‡]Non-standard hyperparameters (see Table 1). *ConvNeXt’s 1×1 pointwise convolution is functionally analogous to MLP but operates within a depthwise-separable structure incompatible with DyT. [§]ConvNeXt-T LN exhibits high inter-seed variance ($\sigma=2.45$ pp). RMT-T’s catastrophic collapse is reported without introducing additional ad-hoc explanatory factors.

Architecture	Topology	Depth / Width	MLP	Δ (pp)	Dominant Factor
DeiT-S	Open (attn)	12 / 384	✓	+6.90	Shallow-wide buffer
DeiT-T	Open (attn)	12 / 192	✓	+3.91	Narrow width
CaiT-XXS24	Open (attn)	26 / 192	✓	−12.34	Deep-narrow compression
ConvNeXt-T [§]	Open (CNN)	—	✓*	−12.11	Structural mismatch
Swin-T	Open (windowed attn)	4-stage / —	✓	−17.18 [‡]	Resolution confound
Vim-T	Closed (SSM)	10 / 384	✓	+3.45	SSM + MLP compensation
PlainMamba-T	Closed (SSM)	12 / 384	—	−0.41	No MLP channel
RWKV-T	Hybrid	12 / —	✓	−2.17	Structural + <code>key_norm</code>
RMT-T	Open (retention)	14 / 64-512	✓	collapse [†]	Retention mech. collapse
Seq2D-S	Closed (BiLSTM)	21 / 192	—	collapse [†]	BiLSTM recurrence

410 We treat the scan-direction hypothesis as suggestive pending validation on additional SSM variants.

411 **Limits of compensation.** PlainMamba-Wide ($d=576$) + DyT collapses deterministically (3/3
412 seeds), while Wide+LN converges to 65.66%. Adding MLP shifts collapse from deterministic to
413 stochastic (seed 42 collapses at epoch 42; seed 123 completes 100 epochs at 63.1%). Near-matched
414 capacity (PlainMamba+MLP ~ 26.3 M vs. PlainMamba-Wide ~ 26.4 M) confirms MLP’s structural
415 role is distinct from capacity.

416 5.4 Factor 3: Topology $\times$ Mechanism Interaction

417 Table 5 summarizes the three-factor analysis. The framework is consistent with six of ten cases.
418 ConvNeXt-T’s depthwise-separable structure produces extreme rank collapse (−51.4%) despite
419 moderate α (0.42), indicating geometric incompatibility outside the α –saturation pathway. RMT-T’s
420 collapse is reported without additional ad-hoc factors. RWKV-T decomposes into structural ($\sim 65\%$)
421 and `key_norm` ($\sim 35\%$) components (Table 2).

422 Topology shapes how tanh compression propagates. In open-loop attention, compression affects
423 per-layer inputs without cross-layer state accumulation; shallow-wide architectures absorb this.
424 In closed-loop SSMs, recursive state propagation carries compression across positions, making
425 MLP compensation critical. The hybrid RWKV-T introduces a third mode via `key_norm`-specific
426 vulnerability. The α -stability window is itself topology-dependent (§4.6): DeiT-S converges for
427 $\alpha \leq 0.50$, RWKV-T at all tested values, Vim-T only at $\alpha=0.50$, while Swin-T and CaiT-XXS24
428 converge broadly but remain ≥ 6.6 – 9.8 pp below LN at their best α .

429 5.5 Mechanism Integration: The α –Saturation–Rank Chain

430 This correlational analysis uses converged checkpoints from four architectures (seed = 42). Two
431 pathways emerge (Figures 5, 3): (i) low learned α (DeiT-S: 0.25, Vim-T: 0.29) keeps tanh near-linear,
432 preserving rank (−3.8% and +12.9%); (ii) high α (RWKV-T: 1.17) drives saturation, compressing
433 rank by −13.8%. Vim-T uniquely *increases* rank under DyT, suggesting its selective scan redistributes
434 rather than loses information. Rank compression is not beneficial regularization: DeiT-T shows larger
435 compression (−27.7%) than DeiT-S (−3.8%) yet smaller accuracy gain (+3.91 vs. +6.90 pp).

436 ConvNeXt-T illustrates a distinct failure: despite moderate α (0.42), its depthwise-separable
437 convolution produces −51.4% rank collapse, indicating structural mismatch outside the α –saturation
438 pathway.

439 **Gradient saturation.** Gradient-level analysis (Appendix P) confirms the compression mechanism:
440 learned α increases with depth and inverse width, driving saturation fractions from 3.9% (Vim-T) to

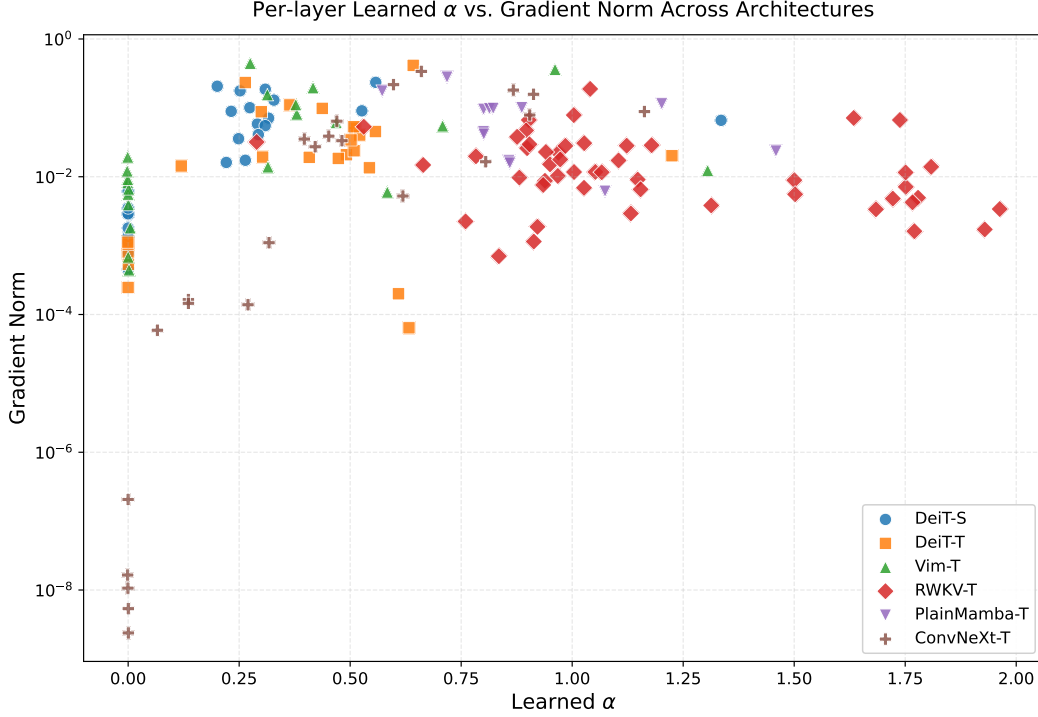


Figure 5: Learned α vs. tanh saturation fraction across architectures (seed = 42). Higher α drives greater saturation, which correlates with effective rank compression (Figure 3).

441 68.3% (CaiT-XXS24), consistent with the depth–width account above.

442 DyT-induced training failures partition into three modes (learn-then-collapse, never-learned,
443 seed-dependent); full analysis is in Appendix O.

444 5.6 Limitations

445 Our experiments use 32×32 resolution; our ImageNet-100 experiments (§K) show DyT effectiveness
446 degrades at 224×224 , establishing a resolution boundary. The 2–3 seed protocol establishes
447 directional trends but precludes formal statistical testing. RMT-T is single-seed; its effect size
448 (collapse) substantially exceeds inter-seed variance in multi-seed conditions, but replication is needed.
449 CaiT-XXS24’s 3-seed results ($\Delta = -12.34$ pp, direction-consistent across all seeds) confirm the single-
450 seed finding. Swin-T uses non-standard hyperparameters, limiting comparability. Vim-T exhibits seed-
451 dependent sign reversal on Tiny-ImageNet (1/3 seeds negative; Δ range $[-19.8, +4.7]$ pp), limiting
452 the robustness of the closed-loop positive conclusion. The catastrophic failure in one seed (val_acc
453 21.4% vs. LN 41.2%) resembles the α -sensitivity pattern observed in other architectures (§4.6),
454 suggesting the DyT benefit in bidirectional SSMs is real but narrowly conditioned on initialization—
455 consistent with Vim-T requiring $\alpha=0.5$ to converge on CIFAR-100.

456 **Three-factor framework.** The framework is post-hoc and correlational, not predictive; with
457 three factors and four architecture-specific exceptions, effective degrees of freedom approach the
458 number of tested architectures. Key single-seed claims—CaiT depth threshold (12L $\rightarrow$ 26L reverses
459 sign), PlainMamba $\pm$ MLP ablations—should be treated as hypotheses requiring replication. The
460 MLP compensation claim rests on one ablation pair (Vim-T $\pm$ MLP) with a capacity-confounded
461 counterexample. No intervention experiments (e.g., fixed α) were performed; the analyses are
462 correlational.

463 **Scope of claims.** The positive DyT effect is specific to *global standard self-attention at moderate*
464 *depth–width ratio*. Four open-loop counter-examples (CaiT, Swin-T, ConvNeXt-T, RMT-T) demon-

strate no other tested mechanism benefits. Swin-T’s deficit is noteworthy given positive ImageNet results from Zhu et al. [2025]; the resolution interaction remains open. Positive results are concentrated in the DeiT family; validation on other architectures (e.g., ViT-B) is needed. The α sweep covers five architectures; ConvNeXt-T was not explored with alternative hyperparameters.

The SSM analysis rests on two architectures; intermediate variants (VMamba’s 4-directional scan) and non-Mamba SSMs (S4, S5) are untested. Our framework predicts VMamba positive and pure-unidirectional S4 neutral-to-negative, but these remain unvalidated.

Additional analyses of scale confounds (Appendix G), baseline accuracy confounds (Appendix H), corruption robustness (Appendix I), computational cost (Appendix J), resolution ceiling effects (Appendix K), and small-scale experiments (Appendix L) are in the supplementary material.

6 Conclusion

We investigated whether DyT can universally replace LayerNorm across architectures with different information-flow topologies. Our experiments on CIFAR-100 and Tiny-ImageNet provide evidence that it cannot. Global standard self-attention gains substantially (DeiT-S: +6.90 pp, DeiT-T: +3.91 pp), confirmed on Tiny-ImageNet (+6.35 pp for DeiT-S). However, this benefit is specific: other open-loop mechanisms (CaiT-XXS24: −12.3 pp; Swin-T: −17.18 pp; ConvNeXt-T: −12.11 pp; RMT-T: collapse) all degrade. Closed-loop SSMs split by MLP availability: Vim-T (+3.45 pp with MLP) vs. PlainMamba-T (−0.41 pp without, depth–width matched), yielding a 7.31 pp gap consistent with MLP as a contributing factor. The hybrid RWKV-T degrades by 2.17 pp—largely recoverable through α tuning (−0.21 pp at $\alpha=1.0$)—with ablation attributing $\sim 65\%$ to structural norms and $\sim 35\%$ to key_norm. The α -stability range is topology-dependent: RWKV-T converges at all tested values, DeiT-S at $\alpha \leq 0.50$, Vim-T only at $\alpha=0.5$. At 224×224 (ImageNet-100), DyT degrades further (18–23 pp below LN), establishing a resolution-dependent failure mode (§K).

These findings yield a three-factor framework—depth–width ratio, MLP availability, and information-flow topology—that is correlational rather than predictive, with four of ten architectures requiring additional considerations. Normalization-free operation via DyT benefits from multi-factor assessment rather than blanket replacement.

Future Work. Validating on ImageNet at higher resolutions, extending to additional architectures (xLSTM, Mamba-2, NLP decoders), increasing seed counts for closed-loop topologies, and developing topology-aware selective replacement strategies are natural next steps.

Broader Impact. Our findings provide topology-aware guidance for practitioners. As an empirical architectural analysis using public benchmarks, we do not foresee negative societal consequences.

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625 A Convergence Speed Analysis

Table 6: Convergence speed and best validation accuracy on CIFAR-100 (100 epochs, mean $\pm$ std across seeds). Epoch $\rightarrow$ 50% / 90%: first epoch reaching 50% / 90% of that run’s best accuracy. Best Epoch: epoch at which best accuracy is achieved.

Architecture	Norm	Seeds	Best Acc (%)	Epoch $\rightarrow$ 50%	Epoch $\rightarrow$ 90%	Best Epoch
DeiT-S	LN	3	58.0 $\pm$ 1.6	19.0 $\pm$ 2.6	53.0 $\pm$ 5.3	91.0 $\pm$ 0.0
DeiT-S	DyT	3	64.9 $\pm$ 0.6	10.0 $\pm$ 1.0	47.7 $\pm$ 1.2	92.0 $\pm$ 3.6
DeiT-T	LN	3	56.9 $\pm$ 0.1	14.7 $\pm$ 1.5	55.7 $\pm$ 3.2	94.0 $\pm$ 1.0
DeiT-T	DyT	3	60.8 $\pm$ 0.6	12.0 $\pm$ 1.0	50.0 $\pm$ 3.5	95.0 $\pm$ 6.2
Vim-T	LN	3	61.1 $\pm$ 0.6	6.7 $\pm$ 1.2	38.7 $\pm$ 1.5	91.0 $\pm$ 5.3
Vim-T	DyT	3	64.5 $\pm$ 0.8	7.3 $\pm$ 0.6	35.7 $\pm$ 1.5	91.7 $\pm$ 2.5
RWKV-T	LN	3	74.6 $\pm$ 0.2	5.0 $\pm$ 0.0	31.0 $\pm$ 0.0	92.3 $\pm$ 3.1
RWKV-T	DyT	3	72.5 $\pm$ 0.2	7.0 $\pm$ 0.0	36.0 $\pm$ 1.0	93.3 $\pm$ 4.5

626 Table 6 jointly reports convergence speed and peak accuracy. DyT improves best accuracy for
627 both open-loop attention architectures and the closed-loop Vim-T, but *degrades* it for the hybrid
628 RWKV-T, consistent with the main results (Table 1).

629 Despite DyT’s faster early convergence—DeiT-S with DyT reaches 50% accuracy nearly twice as
630 fast as LN (10.0 $\pm$ 1.0 vs 19.0 $\pm$ 2.6 epochs) and the 90% threshold is also reached earlier (47.7 $\pm$ 1.2
631 vs 53.0 $\pm$ 5.3)—the epoch at which best accuracy is achieved (Best Epoch) is uniformly late across
632 all conditions ($\sim$ 89–99), showing no systematic difference between normalizations. This indicates
633 that DyT accelerates the *trajectory* through the loss landscape without shifting the point of peak
634 generalization, which is largely governed by the cosine learning rate schedule.

635 For RWKV-T, DyT slows convergence at both thresholds (7.0 vs 5.0 for 50%; 36.0 vs 31.0
636 for 90%) while also lowering peak accuracy, reinforcing the finding that the effect of replacing
637 LayerNorm with DyT is topology-dependent, with hybrid architectures showing degradation.

638 B Throughput Benchmark

Table 7: Inference throughput and latency on a single NVIDIA RTX 5090 GPU. All models are evaluated with batch size 128, 32×32 input resolution, FP32 precision. Each measurement averages 500 timed iterations after 100 warmup iterations. Δ Throughput reports the relative change when replacing LN with DyT.

Architecture	Norm	Throughput (img/s)	Latency (ms/batch)	Δ Throughput (%)
DeiT-T	LN	34 760	3.68	—
DeiT-T	DyT	33 349	3.84	−4.1
DeiT-S	LN	12 186	10.50	—
DeiT-S	DyT	11 713	10.93	−3.9
RWKV-T	LN	13 817	9.26	—
RWKV-T	DyT	11 745	10.90	−15.0
Vim-T	LN	9 522	13.44	—
Vim-T	DyT	9 277	13.80	−2.6

639 DyT introduces a modest throughput reduction across all architectures compared to LN. The open-
640 loop DeiT variants and the closed-loop Vim-T show relatively small slowdowns (2.6–4.1%), whereas
641 RWKV-T exhibits a substantially larger degradation (−15.0%). The disproportionate overhead in
642 RWKV-T likely stems from replacing `key_norm` with DyT, which introduces additional element-wise
643 operations in the recurrent key projection path that are absent in the standard LayerNorm replacement
644 sites of the other architectures.

Table 8: Efficiency comparison between LayerNorm (LN) and Dynamic Tanh (DyT) across architectures grouped by information-flow topology. Throughput measured on a single RTX 5090 with batch size 128 and mixed precision (AMP). Inference memory profiled at batch size 128 in evaluation mode. Relative changes of DyT w.r.t. LN are shown in parentheses.

Topology	Model	Norm	Params (M)	GFLOPs	Throughput (img/s)	Inf. Mem (MB)	Train Mem (MB)
Open-loop	DeiT-T	LN	5.4	0.346	4112.1	109.4	1296.9
		DyT	5.4	0.346	4551.5 ($\uparrow 10.7\%$)	109.4	1441.7 ($\uparrow 11.2\%$)
	DeiT-S	LN	21.3	1.381	3803.7	272.6	2698.1
		DyT	21.3	1.381	3014.3 ($\downarrow 20.8\%$)	272.6	2990.0 ($\uparrow 10.8\%$)
Closed-loop	Vim-T	LN	21.6	0.770	2499.6	317.5	2716.3
		DyT	21.6	0.768	2606.5 ($\uparrow 4.3\%$)	406.1 ($\uparrow 27.9\%$)	2951.9 ($\uparrow 8.7\%$)
Hybrid	RWKV-T	LN	5.8	0.373	2023.8	147.8	2369.6
		DyT	5.8	0.369	1816.5 ($\downarrow 10.2\%$)	171.1 ($\uparrow 15.8\%$)	2924.7 ($\uparrow 23.4\%$)

Table 8 reports training-time efficiency alongside inference metrics. Training memory increases consistently across all architectures (+8.7% to +23.4%), primarily due to caching tanh intermediate values required for gradient computation during backpropagation. Training throughput exhibits no consistent direction: DeiT-T and Vim-T are slightly faster under DyT (+10.7% and +4.3%), while DeiT-S and RWKV-T experience slowdowns (−20.8% and −10.2%). The asymmetry likely reflects architecture-specific interactions between the DyT computation graph and GPU kernel scheduling under mixed precision.

These overheads are acceptable at our experimental scale (small models, 32×32 inputs, single GPU) and do not materially affect research iteration speed. However, the consistent memory increase and variable throughput impact should be considered when deploying DyT in larger-scale training pipelines where memory headroom or throughput constraints are binding.

C CIFAR-100-C Corruption Robustness

Table 9: Mean corruption accuracy (%) on CIFAR-100-C [Hendrycks and Dietterich, 2019] at 32×32 resolution (3-seed mean, seeds 42, 123, and 0). All 15 standard corruption types are averaged across severity levels 1–5. Δ reports the difference DyT − LN. Clean Δ reproduces the CIFAR-100 accuracy difference from Table 1 for comparison.

Architecture	Topology	LN Mean Corr. Acc (%)	DyT Mean Corr. Acc (%)	Δ (pp)	Clean Δ (pp)
DeiT-S	open-loop	43.48	47.83	+4.35	+6.90
DeiT-T	open-loop	41.97	46.16	+4.19	+3.91
Vim-T	closed-loop	41.42	48.21	+6.79	+3.45
RWKV-T	hybrid	54.80	57.53	+2.73	−2.17

DyT improves mean corruption accuracy across all four core architectures, with the largest gain for the closed-loop Vim-T (+6.79 pp)—nearly double its clean-accuracy improvement (+3.45 pp). RWKV-T is the most striking case: despite a −2.17 pp clean-accuracy deficit, DyT still yields a +2.73 pp corruption-robustness gain, suggesting an implicit regularization effect that benefits out-of-distribution generalization independently of in-distribution performance.

Table 10 provides the full per-corruption breakdown, grouped by corruption category.

The per-corruption breakdown reveals distinct patterns across topologies. *Noise corruptions* drive the largest gains uniformly: all three architectures improve by +3.27 to +12.51 pp, with Vim-T and RWKV-T exceeding +8 pp on all three noise types. Vim-T is the only architecture where DyT improves accuracy on every single corruption type, consistent with the hypothesis that closed-loop (recurrent) architectures benefit most from the bounded activation range of DyT.

RWKV-T presents a more nuanced picture. While its noise and glass-blur gains are the largest

Table 10: Per-corruption accuracy (%) on CIFAR-100-C (2-seed mean over seeds 42 and 123; severity levels 1–5 averaged per corruption type). Summary table (Table 9) reports 3-seed means. Corruption types are grouped into four categories: Noise, Blur, Weather, and Digital. $\Delta = \text{DyT} - \text{LN}$; positive values (**bold**) indicate DyT outperforms LN. The final row reports the mean across all 15 corruption types.

Corruption	DeiT-S (<i>open-loop</i>)			Vim-T (<i>closed-loop</i>)			RWKV-T (<i>hybrid</i>)		
	LN	DyT	Δ	LN	DyT	Δ	LN	DyT	Δ
<i>Noise</i>									
Gaussian	30.41	33.77	+3.36	25.19	36.82	+11.63	35.11	47.62	+12.51
Shot	36.41	40.06	+3.65	32.07	43.49	+11.42	44.15	54.08	+9.93
Impulse	31.92	35.19	+3.27	24.03	32.79	+8.76	32.26	40.53	+8.27
<i>Blur</i>									
Defocus	49.53	55.04	+5.51	49.51	54.09	+4.58	61.68	63.45	+1.77
Glass	42.70	42.02	−0.68	31.36	41.31	+9.95	37.48	52.09	+14.61
Motion	46.88	51.31	+4.43	45.14	48.90	+3.76	58.19	58.16	−0.03
Zoom	47.02	52.34	+5.32	45.41	50.71	+5.30	58.49	60.34	+1.85
<i>Weather</i>									
Snow	45.30	51.34	+6.04	44.16	50.98	+6.82	59.87	61.26	+1.39
Frost	42.11	48.73	+6.62	41.38	48.72	+7.34	58.91	60.10	+1.19
Fog	45.18	49.93	+4.75	46.78	48.52	+1.74	63.00	57.45	−5.55
Brightness	50.61	58.56	+7.95	53.71	58.58	+4.87	69.42	67.46	−1.96
<i>Digital</i>									
Contrast	32.26	38.42	+6.16	33.79	36.19	+2.40	50.54	48.23	−2.31
Elastic	51.68	56.45	+4.77	51.19	55.29	+4.10	64.23	64.24	+0.01
Pixelate	52.42	53.67	+1.25	53.80	56.45	+2.65	64.52	62.08	−2.44
JPEG	49.22	52.98	+3.76	47.73	55.93	+8.20	61.14	64.82	+3.68
Mean	43.58	47.99	+4.41	41.68	47.92	+6.24	54.60	57.46	+2.86

in the table (+14.61 pp on glass blur), it degrades on fog (−5.55 pp), contrast (−2.31 pp), pixelate (−2.44 pp), and brightness (−1.96 pp). These are corruptions that primarily affect global intensity or low-frequency structure, suggesting that DyT’s bounded activation range may clip informative signals in the hybrid architecture’s recurrent pathway when the corruption shifts the overall input distribution rather than adding local noise.

D ImageNet-100 Resolution Probe

To probe DyT behavior at higher resolution, we conduct an exploratory mechanism probe on ImageNet-100 [Deng et al., 2009] (224×224 , 100 classes) using DeiT-S—the architecture where DyT achieves its largest gain at 32×32 . These experiments are designed to test whether the capacity limitation analyzed in §K manifests across different α initializations, rather than to establish a competitive baseline.

Setup.

Architecture: DeiT-S (22M parameters, open-loop)

Dataset: ImageNet-100, 224×224 , 100 classes

α initializations: 0.1, 0.5, 1.0

LN baseline: 70.6% (in100-deit-s-ln)

Training: 100 epochs, batch size 64, seed 42, augmented recipe (mixup 0.8, cutmix 1.0, label smoothing 0.1, drop path 0.1)

Results. Table 11 confirms the capacity limitation across three α initializations. Dead-zone duration and final accuracy both improve monotonically with lower α ($22 \rightarrow 16 \rightarrow 0$ dead-zone epochs;

Table 11: ImageNet-100 (224×224) resolution probe: DeiT-S with DyT at varying α_{init} vs. LN baseline. Dead zone denotes the number of initial epochs during which validation accuracy remains near chance ($\sim 1\%$). This is an exploratory mechanism probe; no tested α value approaches the LN baseline. Lower α yields monotonically better performance, but even the best ($\alpha = 0.1$, 52.3%) remains -18.3 pp below LN.

Configuration	Dead zone (epochs)	Val acc @ep50 (%)	Best val acc (%)	Δ vs LN (pp)
LN baseline	—	—	70.6	—
$\alpha_{\text{init}} = 0.1$	0	39.7	52.3	-18.3
$\alpha_{\text{init}} = 0.5$	16	~ 33	50.2	-20.4
$\alpha_{\text{init}} = 1.0$	22	~ 1	48.0	-22.6

689 48.0% $\rightarrow$ 50.2% $\rightarrow$ 52.3% for $\alpha = 1.0, 0.5, 0.1$), but even the best configuration ($\alpha_{\text{init}} = 0.1$,
690 52.3%) remains -18.3 pp below the LN baseline, confirming insufficient normalization capacity
691 rather than a tuning problem. The mechanism underlying these results is analyzed in §K.

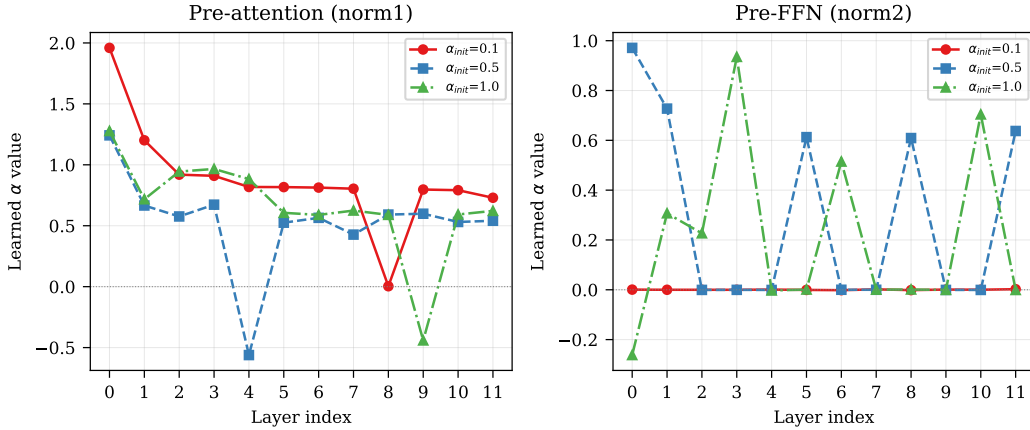


Figure 6: Per-layer learned α values after 100 epochs of training on ImageNet-100 (224×224) for $\alpha_{\text{init}} \in \{0.1, 0.5, 1.0\}$. Left: pre-attention (norm1) α values remain large and positive (0.7–2.0), indicating active scaling. Right: pre-FFN (norm2) α values collapse to near-zero for $\alpha_{\text{init}}=0.1$, while remaining non-negligible for larger initializations, suggesting the FFN path effectively bypasses DyT normalization—a structural asymmetry that may contribute to the capacity limitation observed at 224×224 resolution.

692 E ConvNeXt-T Probe Experiment

693 To test the boundary of our topology-based framework, we apply the same universal DyT replacement
694 protocol to ConvNeXt-T [Liu et al., 2022]—a purely convolutional architecture that is topologically
695 open-loop (no recurrent state) but does *not* share the pre-norm residual pattern characteristic of the
696 Transformer-based architectures in our main study. ConvNeXt-T uses LayerNorm positioned post-
697 depthwise-convolution within inverted bottleneck blocks across a hierarchical multi-stage structure,
698 serving feature-scale management rather than pre-norm training stabilization.

699 **Setup.** We replace all 23 LayerNorm layers in ConvNeXt-T (~ 27.9 M parameters) with DyT
700 ($\alpha_{\text{init}} = 0.5$) and train on CIFAR-100 using the same recipe as our main experiments (AdamW, lr
701 $= 1 \times 10^{-3}$, cosine schedule, 100 epochs, batch size 128). All three seeds (42, 123, 0) are run on a
702 single server (westd-32156) with identical codebase to ensure comparability.

703 **Results.** DyT replacement produces severe degradation across all three seeds: LN achieves $55.10 \pm$
704 2.45% (seeds: 55.2, 57.5, 52.6%) while DyT achieves $42.99 \pm 0.73\%$ (seeds: 43.66, 43.10, 42.20%),

yielding $\Delta = -12.11$ pp. Notably, LN exhibits unusually high inter-seed variance ($\sigma=2.45$ pp), substantially above other architectures ($\sigma \leq 1.55$ pp), while DyT variance is low ($\sigma=0.73$ pp). This confirms that the topology-dependent pattern (open-loop $\Rightarrow$ DyT beneficial) does not extend to architectures outside the pre-norm residual pattern; the architectural role of normalization, not topology alone, determines DyT compatibility. See §5.6 for discussion of architectural coverage limitations.

F PlainMamba-T: Closed-Loop Counter-Example

To test whether the closed-loop topology universally benefits from DyT, we evaluate PlainMamba-T [Yang et al., 2024]—a simplified Mamba variant without bidirectional scanning—on CIFAR-100 using the same protocol as our main experiments.

Setup. PlainMamba-T (~ 12.1 M parameters) uses unidirectional selective scan, making it topologically closed-loop like Vim-T. We train with seeds $\{42, 123, 0\}$ using our standard recipe (AdamW, $\text{lr} = 1 \times 10^{-3}$, cosine schedule, 100 epochs, batch size 128).

Table 12: PlainMamba-T on CIFAR-100: per-seed and aggregate results. $\Delta = \text{DyT} - \text{LN}$.

Seed	LN (%)	DyT (%)	Δ (pp)
42	74.13	73.60	−0.53
123	74.10	73.90	−0.20
0	74.30	73.80	−0.50
Mean $\pm$ std	74.18 $\pm$ 0.11	73.77 $\pm$ 0.15	−0.41 $\pm$ 0.18

Results. Across three seeds, PlainMamba-T shows a consistent DyT degradation of -0.41 ± 0.18 pp (Table 12). All three seeds yield negative Δ values ($-0.53, -0.20, -0.50$ pp), confirming that the effect is directionally robust. LN accuracy is remarkably stable ($74.18 \pm 0.11\%$), with a total spread of only 0.20 pp across seeds. Unlike Vim-T ($+3.45$ pp, 3-seed mean), PlainMamba-T shows DyT *degradation* rather than improvement, demonstrating that the closed-loop topology does not uniformly predict DyT benefit.

Interpretation. The discrepancy between Vim-T and PlainMamba-T under DyT may reflect differences in scan pattern (bidirectional vs. unidirectional), internal architecture details, or parameter scale (22M vs. 12.1M). This finding reinforces the N=1-per-topology limitation discussed in §5.6 and motivates treating topology as an organizing lens rather than a deterministic predictor of DyT effectiveness.

G Scale Confound and Controlled Comparison

A potential confound is model scale. Our architectures form two near-parameter-matched pairs that control for this: DeiT-S (~ 22 M, open-loop) / Vim-T (~ 22 M, closed-loop) and DeiT-T (~ 5.4 M, open-loop) / RWKV-T (~ 5.8 M, hybrid). At the ~ 22 M scale, both architectures benefit from DyT, but the open-loop DeiT-S gains twice as much as the closed-loop Vim-T (the largest gain vs. $+3.45$ pp; Table 1). At the ~ 5.5 M scale, the effect reverses sign entirely: open-loop DeiT-T gains $+3.91$ pp while hybrid RWKV-T loses -2.17 pp. Scale cannot explain a sign reversal at matched capacity, nor a $2\times$ magnitude difference at another matched capacity. Topology is the only factor in our study that consistently covaries with both the direction and magnitude of the DyT effect.

All models are trained from scratch without pretrained weights, producing lower absolute accuracies than transfer-learning baselines. Our DeiT-S result (64.9%) is consistent with prior from-scratch benchmarks: Li et al. [2022] report 65.81% at 200 epochs, and Li et al. [2023] report DeiT-Tiny at 65.08% (200 epochs). Crucially, our study is a *controlled comparison* with normalization type as the only variable; the quantities of interest are the accuracy *differences* (Δ), not absolute values.

H Baseline Accuracy as Confound

One might argue that the decreasing DyT benefit simply reflects a ceiling effect: models with higher baseline accuracy have less room to improve. DeiT-T provides a direct counterexample. Its LN baseline (56.88%) is *lower* than that of DeiT-S (58.0%), yet its DyT gain is also smaller (+3.91 pp vs. DeiT-S’s larger improvement). Under the ceiling-effect hypothesis, DeiT-T’s lower starting accuracy should yield a *larger* Δ , not a smaller one; the observed pattern runs in the opposite direction.

More critically, baseline differences can at most modulate the *magnitude* of improvement; they cannot explain a *sign reversal*. RWKV-T holds the highest baseline among our models (74.63%) and is the only architecture where DyT hurts (−2.17 pp), yet Vim-T still benefits (+3.45 pp) despite a substantially higher baseline (61.08%) than DeiT-T (56.88%). If baseline accuracy drove the effect, one would expect a monotonic negative correlation between LN accuracy and Δ ; the four architectures exhibit no such relationship.

Recurrent-state topology offers a more parsimonious account: a single categorical variable—the presence and type of recurrent state—correctly accounts for both the direction of the DyT effect across topologies and the sign reversal in the hybrid case in our low-resolution experiments, whereas baseline accuracy fails on both counts.

I Corruption Robustness Discussion

Beyond clean accuracy, DyT improves corruption robustness on CIFAR-100-C (32×32): DyT yields higher mean corruption accuracy than LN across all four architectures (DeiT-S: +4.35 pp; DeiT-T: +4.19 pp; Vim-T: +6.79 pp; RWKV-T: +2.73 pp; 3-seed means, Table 9). Notably, RWKV-T benefits from DyT under corruption despite its −2.17 pp clean-accuracy deficit, suggesting an implicit regularization effect under distribution shift. The corruption gain does not simply track clean accuracy: Vim-T’s corruption improvement (+6.79 pp) nearly doubles its clean gain (+3.45 pp), whereas DeiT-S shows the reverse pattern (+4.35 pp corruption vs. its larger clean gain), suggesting that the recurrent state in closed-loop architectures may amplify DyT’s regularization benefit under shift. Across all model $\times$ seed evaluations the direction is consistent, though these results are limited to 32×32 resolution.

J Computational Cost Discussion

Replacing LayerNorm with DyT does not come for free. Table 8 reports training throughput and memory consumption measured on an RTX 5090 (batch size 128, 32×32 , AMP). Training memory increases consistently across all architectures (+8.7% to +23.4%), reflecting the additional intermediate activations cached for backpropagation through the tanh nonlinearity. Training speed, however, shows no consistent direction: DeiT-T and Vim-T are slightly faster with DyT (+10.7% and +4.3%, respectively), while DeiT-S and RWKV-T are slower (−20.8% and −10.2%).

Inference throughput also decreases modestly across all architectures (−2.6% to −15.0%; Table 7), with RWKV-T exhibiting the largest overhead due to DyT replacing the specialized `key_norm` operation. These costs represent a trade-off rather than a free improvement: DyT’s accuracy gains in open-loop attention and closed-loop topologies come at a measurable efficiency penalty. In our experimental setting (small models, 32×32 resolution), the overhead is modest and does not materially affect research iteration speed; practitioners considering DyT at larger scales should weigh these costs against accuracy benefits for their specific deployment constraints.

K Resolution Ceiling: Capacity Limitation at 224×224

The α -sensitivity results in §4.6 reveal topology-dependent fragility on CIFAR-100 (32×32). Our ImageNet-100 experiments at 224×224 expose a more fundamental limitation: **all tested α initializations converge to a narrow accuracy band of 48–52%, falling 18–23 pp below the LN baseline (70.6%)**, indicating a fundamental capacity limitation of tanh-based normalization

substitution that is largely independent of α tuning. The three configurations ($\alpha_{\text{init}} \in \{0.1, 0.5, 1.0\}$) produce final accuracies separated by ~ 2 pp steps—a negligible spread relative to the 18–23 pp deficit from LayerNorm.

Dead-lock mechanism at high α . Recall that $\text{DyT}(x) = \gamma \cdot \tanh(\alpha x) + \beta$ (Equation 1). The gradients with respect to the input and scale parameter are:

$$\frac{\partial \text{DyT}}{\partial x} = \gamma \cdot \alpha \cdot \text{sech}^2(\alpha x), \quad (2)$$

$$\frac{\partial \text{DyT}}{\partial \alpha} = \gamma \cdot x \cdot \text{sech}^2(\alpha x). \quad (3)$$

When $|\alpha x| \gtrsim 2$, $\text{sech}^2(\alpha x) \rightarrow 0$, simultaneously suppressing both gradients. This creates a self-reinforcing dead-lock: saturated activations prevent gradient flow through the layer (Eq. 2), and—critically— α itself cannot update to escape saturation because its own gradient shares the same $\text{sech}^2(\alpha x)$ factor (Eq. 3). The parameter is locked at its initial value.

Dead zone as transient effect. Higher α values trigger longer dead zones—22 epochs for $\alpha = 1.0$, 16 for $\alpha = 0.5$, and 0 for $\alpha = 0.1$ —during which validation accuracy remains near chance ($\sim 1\%$). Although dead-zone duration correlates positively with α , this is a transient phenomenon: once training escapes saturation, all configurations converge to the same narrow accuracy band. The ~ 2 pp separation between adjacent α settings in final accuracy confirms that the initial dead zone does not meaningfully affect asymptotic performance; the dominant bottleneck lies elsewhere.

Bounded output range as the core limitation. Reducing α successfully avoids saturation: $\alpha_{\text{init}} = 0.1$ eliminates the dead zone entirely and achieves the highest DyT accuracy (52.3%). However, this best-case operating point still falls -18.3 pp short of LN (70.6%). The bounded output range of $\tanh([-1, 1])$ fundamentally limits the normalization capacity available to each layer. LayerNorm computes per-token statistics that adaptively rescale activations across the full feature dimension; DyT’s pointwise $\tanh(\alpha x)$ compresses all activations into a fixed interval regardless of their distributional properties. At 224×224 resolution, where pre-activation magnitudes are substantially larger than at 32×32 , this capacity gap becomes insurmountable—and no α value can bridge it, as the narrow 48–52% convergence band demonstrates.

Resolution dependence. The onset of dead-lock is controlled by the product $|\alpha_{\text{init}} \cdot x_0|$, where x_0 denotes the pre-activation magnitude at initialization. For a patch embedding with patch size p applied to an image of resolution $r \times r$, the embedding magnitudes scale with p^2 (the number of pixels per patch). At 32×32 with $p = 4$, pre-activation magnitudes are modest, and $\tanh(\alpha x_0)$ operates in a regime where nonlinear compression is active without triggering saturation—an effective working interval exists. At 224×224 with $p = 16$, the $16^2 = 256$ pixels per patch produce substantially larger embedding magnitudes. In this regime, α values large enough to engage the nonlinear compression of $\tanh$ simultaneously trigger saturation, while values small enough to avoid saturation reduce to the near-linear regime where normalization capacity is insufficient. The result is a resolution-dependent capacity ceiling that cannot be resolved by α tuning alone.

Empirical evidence. We probe this limitation on ImageNet-100 at 224×224 with three α initializations (Table 11). All three converge to a narrow 48–52% band, 18–23 pp below the LN baseline (70.6%, in100-deit-s-ln): $\alpha_{\text{init}} = 0.1$ (in100-deit-s-dyt-a01) achieves 52.3% (-18.3 pp) with no dead zone; $\alpha_{\text{init}} = 0.5$ (in100-deit-s-dyt-r2) reaches 50.2% (-20.4 pp) after a 16-epoch dead zone; $\alpha_{\text{init}} = 1.0$ (in100-deit-s-dyt-a10-r2) reaches 48.0% (-22.6 pp) after a 22-epoch dead zone. The uniform ~ 2 pp spacing between adjacent α settings, combined with the consistently large deficit from LN, indicates that α selection modulates performance only marginally within a regime bounded far below LN capacity. By contrast, $\alpha_{\text{init}} = 0.5$ on CIFAR-100 (32×32) yields 64.9% (3-seed mean)—a $+6.90$ pp gain over LN (58.0%)—confirming that the limitation is resolution-specific, not inherent to any particular α value.

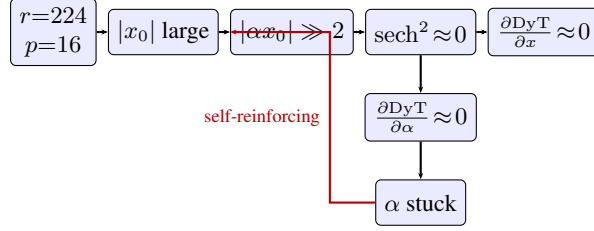


Figure 7: Causal chain of the α dead-lock mechanism at high α . High-resolution inputs produce large pre-activations that saturate $\tanh(\alpha x)$, suppressing gradients for both the signal ($\partial \text{DyT} / \partial x$) and the scale parameter ($\partial \text{DyT} / \partial \alpha$). Because α cannot update, saturation persists—creating a self-reinforcing loop (red arrow). Reducing α breaks this loop but does not recover LN-level performance due to insufficient normalization capacity of the bounded $\tanh$ output (see text).

Scope and interpretation. This capacity limitation is observed in our ImageNet-100 experiment with a single architecture (DeiT-S), one higher-resolution dataset, and three α values; we present it as an exploratory mechanistic probe of DyT’s resolution boundary rather than a general claim about $\tanh$ -based normalization at scale. Our primary conclusions regarding topology-dependent DyT effectiveness are validated in the 32–64 px regime where DyT operates within its effective capacity; extending these findings to higher resolutions remains an open challenge. Whether activation functions with unbounded or adaptive-range outputs can close the gap at 224×224 and beyond is an important direction for future work.

L Small-Scale Experiments as Controlled Testbeds

Our primary experiments operate at 32×32 (CIFAR-100) and 64×64 (Tiny-ImageNet)—substantially below the 224×224 standard in modern vision benchmarks. This design is deliberate: small-scale controlled settings isolate normalization type as the sole variable while minimizing confounders from dataset scale, training cost, and hyperparameter sensitivity. The resulting computational efficiency enables our multi-seed, multi-architecture factorial protocol; replicating the same design at ImageNet scale would require approximately $50\times$ the compute budget, precluding the systematic comparison across four architectures and three seeds that underpins our topology analysis. This experimental strategy has extensive precedent: the original ViT paper [Dosovitskiy et al., 2021] includes CIFAR-10/100 experiments alongside ImageNet results, and studies of training dynamics, normalization effects, and architectural design principles routinely employ small-scale benchmarks to establish controlled findings before scaling.

The mechanistic understanding derived from these experiments extends beyond the specific resolution. Our topology-dependent spectrum—DyT beneficial in open-loop attention architectures, diminishing in closed-loop, harmful in hybrid—identifies recurrent state coupling as a design axis for normalization-free architectures, a structural insight independent of input resolution. Equally important, our ImageNet-100 probe (§K) demonstrates that DyT effectiveness is *not* trivially transferable across resolutions: the α dead-lock and insufficient normalization capacity at 224×224 reveal failure modes that small-scale experiments alone cannot predict. This negative result carries direct practical value—it prevents practitioners from naïvely extrapolating small-scale DyT gains to higher-resolution settings without accounting for the resolution-dependent capacity limitations identified in §K.

M Per-Seed Tiny-ImageNet Results

Table 13 reports individual seed results. Vim-T shows the largest per-seed variance (Δ spans -1.19 to $+4.40$ pp), with one sign reversal, aligning with its narrow α -stability window (§4.6). DeiT-S and RWKV-T show consistent sign across all seeds. The closed-loop Tiny-ImageNet result should be characterized as marginal rather than robust.

Table 13: Per-seed Tiny-ImageNet accuracy (%) for DyT vs. LayerNorm. $\Delta = \text{DyT} - \text{LN}$. All architectures include 3 seeds each (42, 123, 0). Per-seed accuracy values are rounded to one decimal place; mean Δ is computed from unrounded values.

Architecture	Seed	DyT	LN	Δ (pp)
DeiT-S (open)	42	46.4	40.6	+5.80
	123	46.1	39.2	+6.90
	0	45.9	39.6	+6.34
	<i>Mean$\pm$std</i>	46.2 ± 0.3	39.8 ± 0.7	+6.35
DeiT-T (open)	42	41.8	41.2	+0.59
	123	42.0	41.7	+0.36
	0	42.5	41.3	+1.12
	<i>Mean$\pm$std</i>	42.1 ± 0.3	41.4 ± 0.3	+0.69
Vim-T (closed)	42	46.9	42.5	+4.40
	123	44.2	45.4	−1.19
	0	42.5	41.5	+1.00
	<i>Mean$\pm$std</i>	44.5 ± 2.2	43.1 ± 2.0	+1.40
RWKV-T (hybrid)	42	53.4	54.9	−1.50
	123	51.6	54.7	−3.10
	0	52.7	54.3	−1.55
	<i>Mean$\pm$std</i>	52.6 ± 0.9	54.6 ± 0.3	−2.05

N Sensitivity to Initialization Scale (α)

O Training Collapse in Closed-Loop DyT

DyT-induced training failures partition into three modes: (1) *learn-then-collapse*—RMT-T reaches $\sim 30\%$ accuracy before collapsing to $\sim 2\%$ at epoch 16; (2) *never-learned*—Sequencer2D-S never exceeds $\sim 5\%$ under DyT ($\Delta = -55.9$ pp); (3) *seed-dependent*—Vim-T on Tiny-ImageNet collapses in 1/5 seeds at epoch 3, entering a deadlock phase (loss $\approx \ln(200)$) before recovering at epoch 50. These modes differ in onset, reversibility, and determinism, suggesting distinct instability mechanisms.

The Vim-T collapse affects 1/5 seeds (20% incidence); the +1.40 pp mean in Table 4 uses the standard 3-seed protocol. Including all five seeds yields -2.19 pp, underscoring the outsized impact of a single catastrophic initialization. This constitutes a tail risk of the closed-loop + DyT combination rather than systemic failure. PlainMamba-Wide + DyT shows a distinct pattern: deterministic collapse (3/3 seeds), shifted to stochastic by adding MLP (§5.3).

P Gradient Saturation Analysis

We examine gradient-level mechanisms using 1000-sample gradient flow measurements from converged checkpoints (seed = 42).

Depth-dependent α growth. Learned α increases with depth and inverse width: DeiT-S ($12L/d=384$) maintains $\alpha = 0.27$; DeiT-T ($12L/d=192$) learns $\alpha = 0.47$; CaiT-XXS24 ($24+2L/d=192$) reaches $\alpha = 0.60$ – 0.85 in deep layers, with the final norm at $\alpha = 6.35$.

The saturation mechanism. High α drives saturation where $\text{sech}^2(\alpha x) \rightarrow 0$ suppresses gradients: CaiT reaches 68.3% saturation, vs. 12.4% for DeiT-S, 10.5% for DeiT-T, and 3.9% for Vim-T. Within CaiT, DyT exhibits $3.21\times$ gradient decay across SA blocks vs. $1.79\times$ under LN ($1.8\times$ steeper), with per-layer DyT/LN ratios declining from 0.49 (SA-0) to 0.27 (SA-23).

Width buffer. DeiT-T learns $\alpha = 0.468$ vs. DeiT-S $\alpha = 0.269$ at matched depth—a $1.74\times$ ratio, providing a gradient-level account for the width effect: wider layers maintain lower α and saturation.

Table 14: α -sensitivity sweep on CIFAR-100. Vim-T (closed-loop) collapses at 4 of 5 tested α values; DeiT-S (open-loop) converges at $\alpha \leq 0.50$, degrades at $\alpha=1.0$, and collapses at $\alpha=2.0$; RWKV-T (hybrid) converges at all tested values. RWKV-T performance improves with higher α , largely closing the DyT-LN gap; Swin-T (windowed attention) peaks near $\alpha=1.0$ but remains 9.8 pp below LN; CaiT-XXS24 (deep self-attention) improves gradually with higher α (45.98–51.33%, basic recipe) but remains ≥ 6.57 pp below the basic-recipe LN baseline across all four tested values. A run is classified as “collapsed” if its best validation accuracy remains at or below chance level ($\leq 3\%$ for CIFAR-100’s 100 classes); “degraded” denotes accuracy well above chance but substantially below the best converged result for that architecture. [§]3-seed mean $\pm$ std (seeds 42, 0, 123); all other entries are single seed (seed = 42). [†]Single-seed probe (see Table 1). [‡]Non-standard hyperparameters.

Architecture	α	Val Acc (%)	Outcome	Exp. ID
Vim-T (closed-loop)	0.25	~ 1	Collapsed	alpha-vim-a025
	0.50	63.7	Converged	alpha-vim-a050
	0.75	~ 2	Collapsed	alpha-vim-a075
	1.00	~ 3	Collapsed	alpha-vim-a100
	2.00	~ 2	Collapsed	alpha-vim-a200
DeiT-S (open-loop)	0.10	61.8	Converged	alpha-deit-a010
	0.25	63.6	Converged	alpha-deit-a025
	0.50	65.1	Converged	alpha-deit-a050
	1.00	~ 30	Degraded	alpha-deit-a100
	2.00	~ 1	Collapsed	alpha-deit-a200
RWKV-T (hybrid)	0.10	61.8	Converged	alpha-rwkv-a010
	0.25	68.7	Converged	alpha-rwkv-a025
	0.50	72.26	Converged	pvtB-1b
	1.00	$74.42 \pm 0.14^{\S}$	Converged	alpha-rwkv-a100
	2.00	74.09	Converged	alpha-rwkv-a200
Swin-T ^{†‡} (windowed attn)	0.30	40.61	Converged	swin-t-dyt-a03-s42
	0.50	46.73	Converged	swin-t-dyt-s42-r3
	1.00	54.26	Converged	swin-t-dyt-a10-s42
	2.00	53.31	Converged	swin-t-dyt-a20-s42
CaiT-XXS24 [†] (deep attn)	0.30	45.98	Converged	cait-xxs24-dyt-a03-s42
	0.50	49.5	Converged	cait-xxs24-dyt-s42
	1.00	50.7	Converged	cait-xxs24-dyt-a10-s42
	2.00	51.33	Converged	cait-xxs24-dyt-a20-s42

Open question: Vim-T. Vim-T shows the lowest saturation (3.9%) yet steep gradient decay ($8.02\times$ under DyT vs. $3.29\times$ under LN), while still gaining +3.45 pp. The mechanism remains unclear; bidirectional SSM dynamics and MLP compensation are candidate explanations.

Broader Impact

This work is a controlled empirical study of normalization layer alternatives in neural network architectures. We analyze whether Dynamic Tanh (DyT) can replace LayerNorm across architectures with different information-flow topologies (open-loop, closed-loop, hybrid), using standard image classification benchmarks at low resolution.

Potential positive impact. Our findings provide practitioners with topology-aware guidance for when normalization-free techniques can be safely adopted and when they should be avoided. By identifying that DyT effectiveness is topology-dependent rather than universal, we help prevent silent accuracy degradation from blanket adoption of normalization replacement strategies.

Potential negative impact. We do not foresee direct negative societal consequences from this work. The research is a purely architectural analysis that does not introduce new capabilities, datasets, or

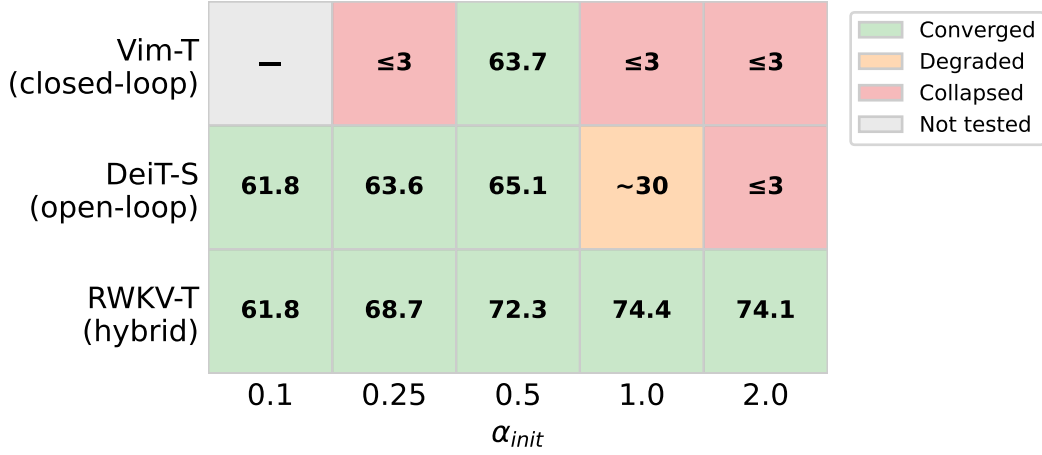


Figure 8: α -sensitivity heatmap across topologies on CIFAR-100. Green cells indicate successful convergence (with accuracy shown); red cells indicate training collapse ($\leq 3\%$ accuracy); gray cells were not tested. Each topology exhibits a distinct α -stability window: RWKV-T (hybrid) converges at all tested values; DeiT-S (open-loop) converges at $\alpha \leq 0.50$, degrades at $\alpha=1.0$, and collapses at $\alpha=2.0$; Vim-T (closed-loop) converges only at $\alpha=0.50$; Swin-T (windowed attention) converges at all four tested values with performance peaking near $\alpha=1.0$; CaiT-XXS24 (deep self-attention) converges at all four tested values with gradual improvement at higher α (basic recipe), but remains ≥ 6.57 pp below the basic-recipe LN baseline. Single seed (seed=42) except where noted.

905 deployment-facing systems. It does not involve human subjects, personally identifiable information,
 906 or dual-use technology.

907 **Computational resources.** All experiments were conducted on single NVIDIA RTX 5090 GPUs.
 908 Across 117 total experiments (97 with recorded compute), the project consumed approximately **125**
 909 **GPU-hours** (102.8 from successful experiments, 14.9 from negative-result experiments, and 7.0
 910 from failed or terminated runs). All training was performed at controlled low-resolution settings
 911 (32×32 and 64×64), resulting in a modest computational footprint. We estimate the total energy
 912 consumption at approximately 50–60 kWh, corresponding to a carbon footprint well under 25 kg
 913 CO₂e depending on the electricity grid.

914 NeurIPS Paper Checklist

915 1. Claims

916 Question: Do the main claims made in the abstract and introduction accurately reflect the
 917 paper’s contributions and scope?

918 Answer: [Yes]

919 Justification: The abstract and introduction claim that topology serves as an organizing lens
 920 for understanding DyT effectiveness patterns within pre-norm residual architectures, which is
 921 directly supported by experiments across ten architectures on CIFAR-100, with cross-dataset
 922 validation on Tiny-ImageNet.

923 2. Limitations

924 Question: Does the paper discuss the limitations of the work performed by the authors?

925 Answer: [Yes]

926 Justification: Section 5.6 discusses dataset scale, single architecture per topology, seed count,
 927 absence of formal significance testing, and restriction to vision classification.

928 **3. Theory assumptions and proofs**

929 Question: For each theoretical result, does the paper provide the full set of assumptions and a
930 complete (or correct) proof?

931 Answer: [N/A]

932 Justification: This is an empirical study with no theoretical results or proofs.

933 **4. Experimental result reproducibility**

934 Question: Does the paper fully disclose all the information needed to reproduce the main
935 experimental results of the paper to the extent that it affects the main claims and/or conclusions
936 of the paper (regardless of whether the code and data are provided or not)?

937 Answer: [Yes]

938 Justification: Section 3 reports all architecture configurations, hyperparameters, training sched-
939 ules, random seeds, and evaluation protocols. Per-seed results are provided in the Appendix.

940 **5. Open access to data and code**

941 Question: Does the paper provide open access to the data and code, with sufficient instructions
942 to faithfully reproduce the main experimental results, as described in supplemental material?

943 Answer: [Yes]

944 Justification: All datasets used (CIFAR-100, Tiny-ImageNet) are publicly available. Code will
945 be released upon publication.

946 **6. Experimental setting/details**

947 Question: Does the paper specify all the training and test details (e.g., data splits, hyperparam-
948 eters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

949 Answer: [Yes]

950 Justification: Section 3 (Method) specifies architectures, optimizer, learning rate schedule, data
951 augmentation, batch size, training epochs, and evaluation protocol. Additional details are in
952 the Appendix.

953 **7. Experiment statistical significance**

954 Question: Does the paper report error bars suitably and correctly defined or other appropriate
955 information about the statistical significance of the experiments?

956 Answer: [Yes]

957 Justification: We report mean $\pm$ std across 2–3 seeds for all experiments and provide per-seed
958 results in the Appendix. We acknowledge in Section 5.6 that the small seed count precludes
959 formal significance testing.

960 **8. Experiments compute resources**

961 Question: For each experiment, does the paper provide sufficient information on the computer
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963 experiments?

964 Answer: [Yes]

965 Justification: The Appendix reports GPU type and approximate training time per experiment.

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974 impacts of the work?

975 Answer: [Yes]

976 Justification: The Conclusion includes a Broader Impact paragraph noting that our findings
977 serve as practitioner guidance and identifying no anticipated negative societal impacts.

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